Paleoproterozoic intraplate magmatism and basin development on the Kaapvaal Craton: Age, paleomagnetism and geochemistry of ~1.93 to ~1.87 Ga post-Waterberg dolerites

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ABSTRACT

We report U–Pb baddeleyite crystallization ages of ~1927 and ~1879 to ~1872 Ma for dolerite sills intruding the Waterberg Group in Botswana and South Africa. These data increase the known extent of ~1.9 Ga intraplate magmatism in southern Africa and place tighter age constraints on the Waterberg Group than previously available. In South Africa, ~1.88 Ga dolerite intrudes upper Waterberg strata, constraining most, if not all, of the succession to have accumulated between ~2.06 Ga (age of the underlying Bushveld Complex) and ~1.88 Ga. This is consistent with derivation of much of the group from uplifted sources in reactivated segments of the Limpopo Belt. The dolerites are typical continental tholeiites, but their trace-element contents discriminate them from dolerite sills of the 1.1 Ga Umkondo Igneous Province, which occur in the same region. Paleomagnetic samples from dolerite intrusions in the Waterberg Group in South Africa (including one sill with a U–Pb baddeleyite age of ~1872 Ma), and from dolerite sills and basalt flows in the Soutpansberg Group to the east-northeast, yield antipodal directions with a site mean pole at 15.6°north, 17.1°east, A_{95} = 8.9°. These data are interpreted to indicate that the ~1879 to ~1872 Ma dolerites were intruded into the Waterberg Group during voluminous magmatism associated with development of the Soutpansberg rift basin. Older, ~1927 Ma dolerite in Botswana is similar in age and geochemistry to basalts in the craton-margin Olifantshoek Supergroup, suggesting that the mafic magmatism in those two regions is genetically related.

Introduction

The Waterberg Group is a thick Proterozoic sedimentary succession deposited on Archean and Paleoproterozoic rocks of the Kaapvaal Craton (Figure 1), including the ~2.06 Ga Bushveld Complex (Walraven and Hattingh, 1993; Walraven, 1997; de Waal and Armstrong, 2000; Buick *et al.*, 2001; Eglington and Armstrong, 2004). Rb–Sr and U–Pb dates of ~1.4 to ~1.3 Ga for the Pilanesberg Dyke Swarm and syenite of the Leeuwfontein complex that intrude parts of the group have generally been inferred to provide a younger age limit for the group (*e.g.* Jansen, 1982; Tankard *et al.*, 1982; Walraven *et al.*, 1990; Emerman, 1991).

Widespread dolerite sills that intrude the Waterberg Group and predate Jurassic Karoo magmatism have commonly been grouped together as "post-Waterberg dolerites" (or diabases), the implicit assumption being that they represent a single magmatic event (*e.g.* du Toit, 1939; Jones and McElhinny, 1966; SACS, 1980; Jansen, 1982). Paleomagnetic and geochronological data

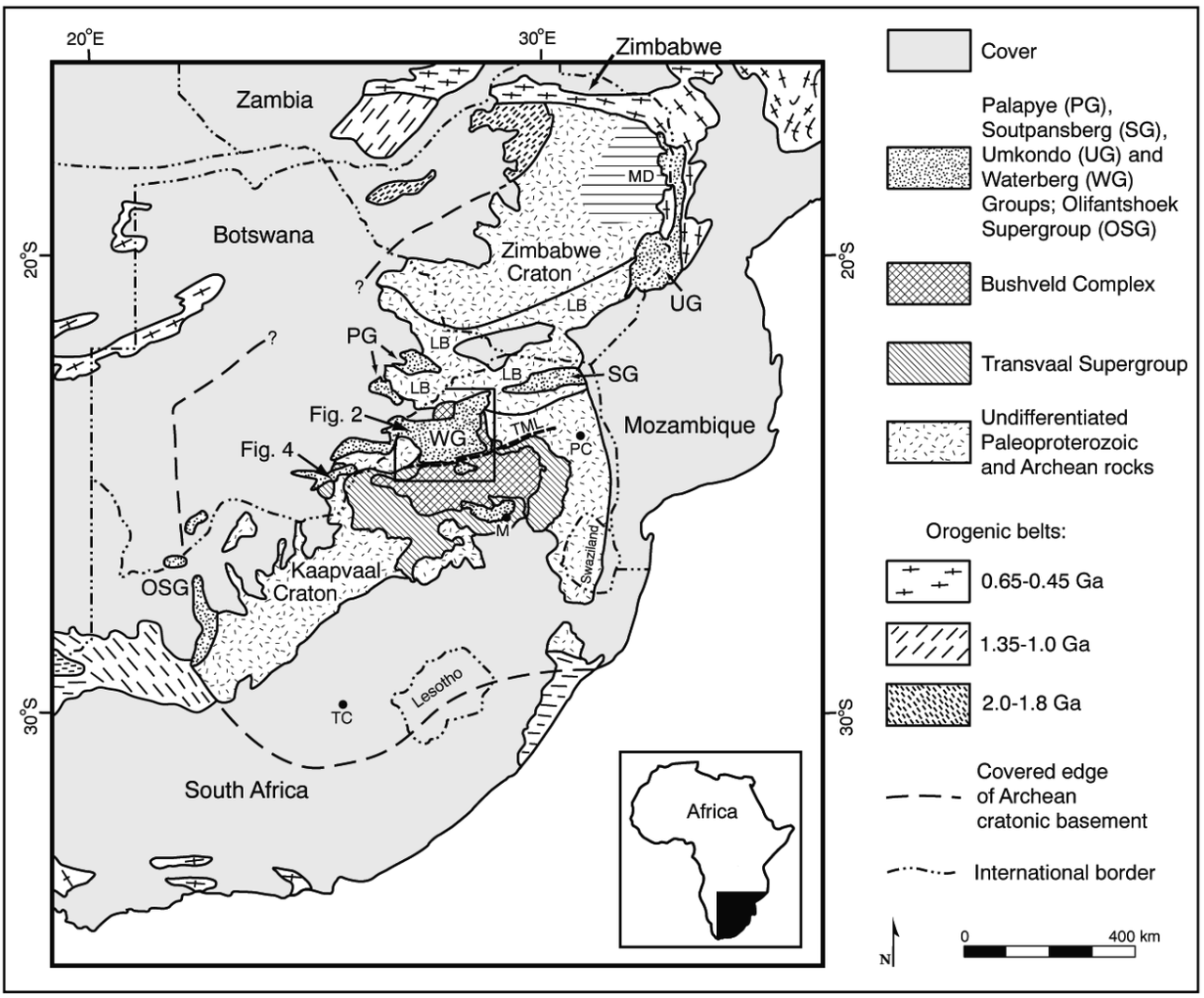


Figure 1. Proterozoic geological framework of the Kaapvaal Craton and adjacent regions, showing extent of post-Bushveld Proterozoic supracrustal successions discussed in text. Other labeled features not defined in legend: LB: Limpopo Belt; M: Middelburg; MD: main extent of Mashonaland dolerite sheets, from Wilson (1990); PC: Phalaborwa Complex; TML: Thabazimbi–Murchison Lineament; TC: Trompsburg Complex. Modified from Zimbabwe Geological Survey (1994), Keyser (1997), Key and Mothibi (1999) and Hanson (2003). The Paleoproterozoic "Kheis Belt" shown along the exposed southwestern margin of the Kaapvaal Craton in previous regional tectonic maps is omitted here, following Eglington and Armstrong (2004).

indicate that many of these intrusions belong to the ~1.1 Ga Umkondo Igneous Province, which extends over much of southern Africa (Jones and McElhinny, 1966; Pancake, 2001; Hanson, 2003). Here we show that some of the post-Waterberg dolerites are considerably older than the Umkondo Province and were emplaced at ~1927 and ~1879 to ~1872 Ma, providing a significantly tighter bracket on the age of the Waterberg Group than previously available. Paleomagnetic data for the ~1879 to ~1872 Ma intrusions yield a new Paleoproterozoic pole for the Kaapvaal Craton, and comparisons between the ~1927 and ~1879 to ~1872 Ma post-Waterberg dolerites and broadly coeval mafic rocks elsewhere in South Africa and Zimbabwe help to constrain the nature and temporal and spatial distribution of intraplate Paleoproterozoic magmatism in the region.

Stratigraphy and depositional setting of the Waterberg Group

The Waterberg Group has been the subject of numerous stratigraphic and sedimentological studies in South Africa, and the following information is primarily from Jansen (1982), Callaghan *et al.* (1991), van der Neut *et al.* (1991) and Callaghan (1993). Across the Waterberg plateau and adjacent areas (Figures 2 and 3), the group has a composite stratigraphic thickness of as much as 7km. The lower part of the group there (Swaershoek, Alma and Sterkrivier Formations), and the entire succession exposed in the Middleburg area, consist dominantly of sandstones and rudites (conglomerates and sedimentary breccias) deposited in alluvial fan and fan-delta environments. These rocks accumulated at least partly in localized basins controlled by movement along the Thabazimbi–Murchison Lineament

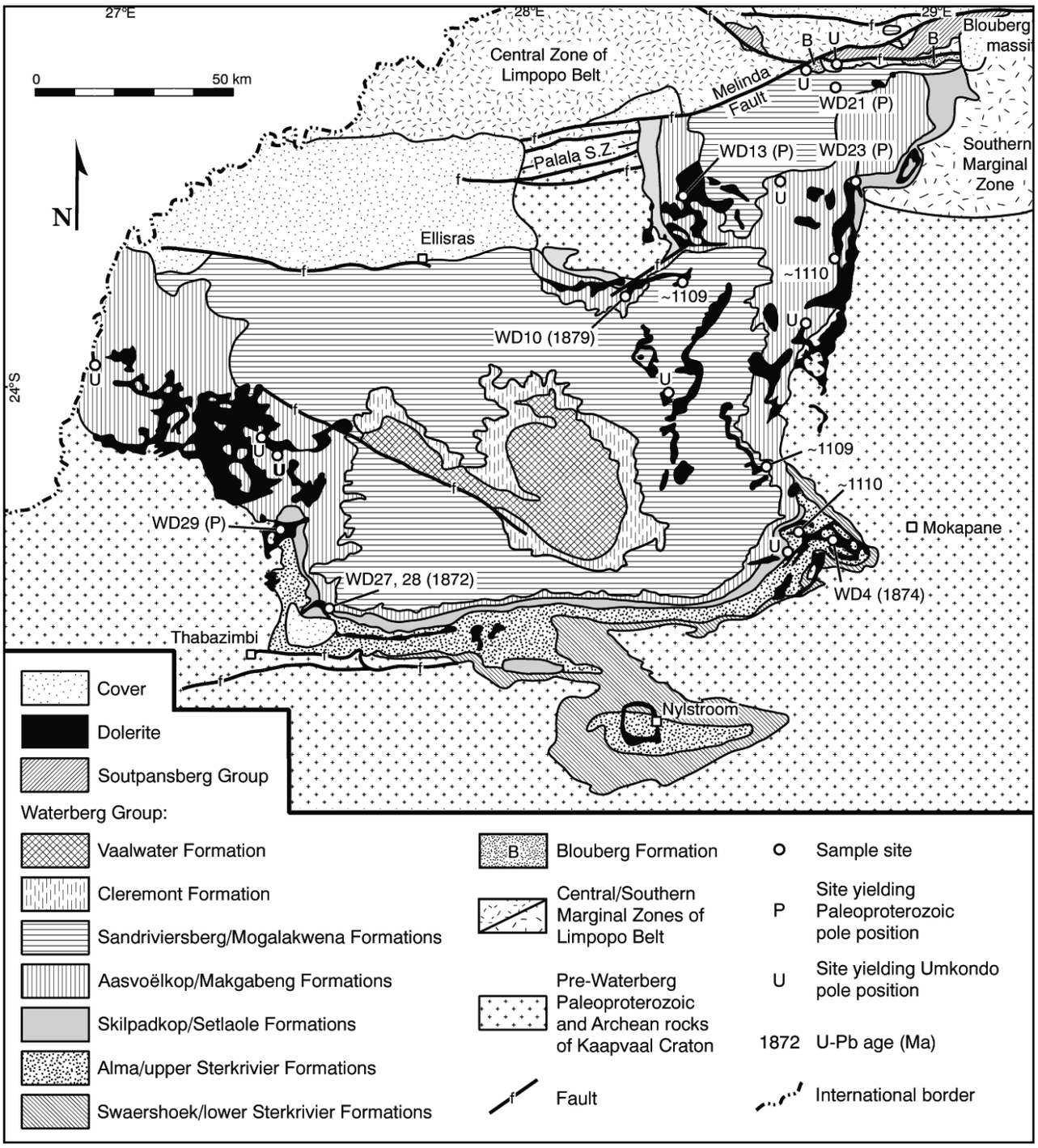


Figure 2. Geological map of the Waterberg Group in Northern Province, South Africa, from Keyser (1997), with modifications in the Blouberg area from Bumby *et al.* (2001a). See Figure 1 for location. Laterally equivalent formations in the Waterberg Group are combined as single map units. Geochronological and paleomagnetic sample localities for Paleoproterozoic dolerite intrusions are indicated (with age in Ma). Sites yielding ~1.1 Ga Umkondo ages or pole positions are also shown, based on authors' unpublished data.

(du Plessis, 1987; Callaghan *et al.*, 1991), a long-lived, lithospheric-scale transcurrent fault zone cutting the Kaapvaal Craton (Figure 1; Good and de Wit, 1997). Significant amounts of felsic lava are intercalated in the Swaershoek and laterally equivalent lower Sterkrivier Formations.

Higher in the succession, coarse fluvial clastics in the Skilpadkop and Setlaole Formations were deposited in narrow braidplains adjacent to uplifted fault blocks along the Thabazimbi–Murchison Lineament and the Melinda Fault Zone to the north (Figure 2). Overlying eolian and lacustrine strata of the Makgabeng and Aasvoëlkop Formations accumulated during more stable conditions. Renewed influx of voluminous coarse clastics formed a vast braidplain that is recorded by sandstones and rudites of the Mogalakwena and Sandriviersberg Formations and undoubtedly extended significantly farther south than the present outcrops of

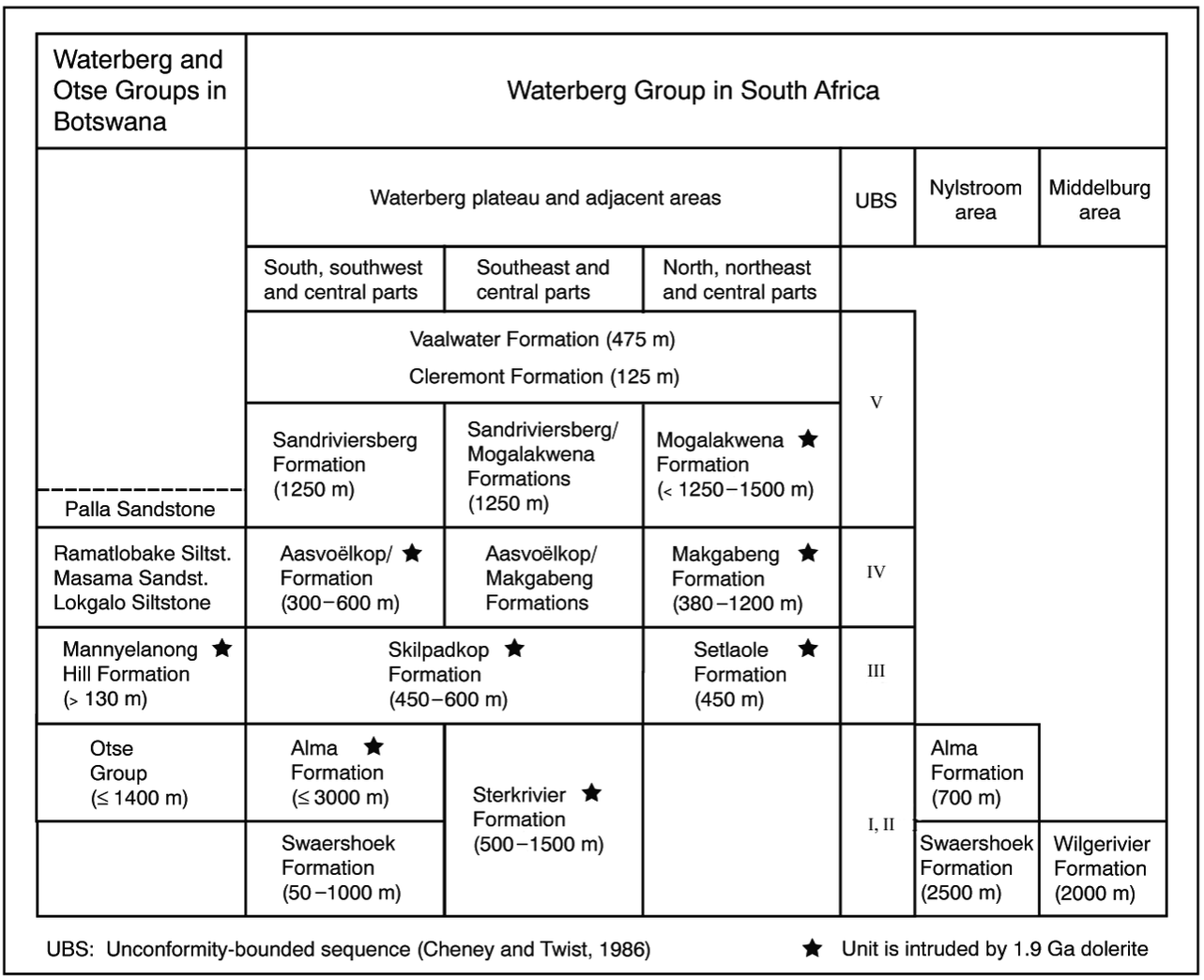


Figure 3. Stratigraphy of the Waterberg Group in South Africa and Botswana, from Callaghan *et al.* (1991) and Carney *et al.* (1994). Thickness of units from Jansen (1982), Key (1983) and Aldiss *et al.* (1989). Total thickness of Waterberg strata above Mannyelanong Formation in Botswana is unknown (Williamson, 1996)

these units. Paleocurrent and lithofacies patterns indicate derivation of the detritus from the east-northeast. Sandstones and mudstones of the overlying Cleremont and Vaalwater Formations are inferred to have accumulated in shallow-marine to littoral settings during a regional transgression that occurred as sediment supply from the east-northeast diminished.

The Waterberg Group is thinner and less well exposed in Botswana. A probable correlative of the Alma Formation there is the Otse Group (Figure 3), which is an areally restricted unit of alluvial fan and more distal fluvial deposits that accumulated on an irregular erosional surface (Key, 1983; Carney *et al.*, 1994). This unit is succeeded by more widespread sandstones and conglomerates of the Mannyelanong Formation, which is the basal unit of the Waterberg Group as defined in Botswana. The Mannyelanong Formation is correlated with the Skilpadkop Formation in South Africa (Carney *et al.*, 1994) and forms discontinuous outcrops in southeastern Botswana that are erosional remnants of a sheet of alluvial fan and braidplain deposits derived from an upfaulted source area to the south (Key, 1983; Aldiss *et al.*, 1989; Williamson, 1996). Near Kanye, these deposits rest unconformably on granite and syenite of the Moshaneng Complex (Figure 4), which has yielded a U–Pb zircon age of 2054 ± 2 Ma and is considered to relate to the much larger Bushveld Complex (Key and Ayres, 2000). The Mannyelanong Formation is succeeded by intertonguing lacustrine, fluvial and fluvial-deltaic deposits correlated with the Aasvoëlkop and lower Sandriviersberg Formations in South Africa (Figure 3; Carney *et al.*, 1994; Williamson, 1996).

Post-Waterberg dolerites

Dolerite occurs as sills, irregular bodies and dykes intruding the Waterberg Group and underlying rocks (*e.g.* Sharpe, 1981; Jansen, 1982; Brandl, 1986; 1996; Carney *et al.*, 1994; Pancake, 2001). The sills are ≤ 120 m thick, typically show fine- to medium-grained diabasic textures (grain size ≤ 2 to 3mm) and comprise plagioclase, augite, pigeonite, magnetite, generally minor enstatite and biotite, interstitial microgranophyre and sparse olivine. Pyroxene shows reaction rims of

Table 1. U-Pb isotopic data

		Concentrations				Isotopic ratios									Ages (Ma)			
Sample (a)	Weight (µg) (b)	U (ppm)	Pb (ppm)	Th/U (c)	Pb_c (pg) (d)	$^{206}Pb/^{204}Pb$ (e)	$^{208}Pb/^{206}Pb$ (f)	$^{206}Pb/^{238}U$ (f)	% err (g)	$^{207}Pb/^{235}U$ (f)	% err (g)	$^{207}Pb/^{206}Pb$ (f)	% err (g)	corr. coef.	$^{206}Pb/^{238}U$ (h)	$^{207}Pb/^{235}U$ (h)	$^{207}Pb/^{206}Pb$ (h)	% disc. (i)
WD04																		
B1	5.1	174	54.4	0.05	6.1	3013	0.015	0.322222	0.08	5.0777	0.10	0.11429	0.05	0.86	1800.6	1832.4	1868.7 ± 0.9	3.6
B2	1.9	302	94.6	0.07	3.9	3076	0.020	0.321069	0.09	5.0591	0.11	0.11428	0.07	0.81	1794.9	1829.3	1868.6 ± 1.2	3.9
B3	2.1	352	108.0	0.13	1.5	9589	0.042	0.308820	0.06	4.8538	0.08	0.11399	0.06	0.75	1734.9	1794.3	1864.0 ± 1.0	6.9
B8	1.1	432	139.7	0.07	1.7	6030	0.020	0.331716	0.06	5.2363	0.08	0.11449	0.05	0.79	1846.7	1858.5	1871.8 ± 0.9	1.3
B9	0.7	411	134.3	0.07	5.3	1181	0.021	0.335095	0.19	5.2951	0.24	0.11461	0.12	0.85	1863.0	1868.1	1873.7 ± 2.2	0.6
WD10																		
B1	2.5	237	77.2	0.05	1.6	7874	0.014	0.336115	0.06	5.3282	0.08	0.11497	0.05	0.73	1868.0	1873.4	1879.4 ± 1.0	0.6
B4	0.7	418	140.5	0.15	1.0	6378	0.044	0.337114	0.09	5.3431	0.12	0.11495	0.08	0.72	1872.8	1875.8	1879.1 ± 1.5	0.3
B5	0.8	512	169.2	0.11	0.9	10395	0.031	0.335764	0.06	5.3191	0.07	0.11490	0.05	0.76	1866.3	1871.9	1878.3 ± 0.9	0.6
B6	0.9	508	167.4	0.08	1.0	9944	0.023	0.336612	0.06	5.3337	0.08	0.11492	0.05	0.77	1870.4	1874.3	1878.6 ± 1.0	0.4
WD27z																		
B1	2.2	326	106.2	0.07	3.0	5150	0.020	0.333938	0.06	5.2695	0.10	0.11445	0.08	0.64	1857.4	1863.9	1871.2 ± 1.4	0.7
B2	2.6	207	63.7	0.03	2.3	4883	0.009	0.318315	0.08	5.0188	0.14	0.11435	0.12	0.53	1781.5	1822.5	1869.7 ± 2.2	4.7
B3	1.6	282	91.4	0.03	0.8	12257	0.008	0.336249	0.09	5.3120	0.20	0.11458	0.18	0.47	1868.6	1870.8	1873.2 ± 3.2	0.2
B6	2.3	211	65.9	0.07	1.1	9364	0.022	0.319545	0.07	5.0364	0.09	0.11431	0.06	0.73	1787.5	1825.5	1869.0 ± 1.1	4.4
JP8b																		
B1	1.4	265	91.7	0.15	2.3	3586	0.043	0.346121	0.08	5.6364	0.11	0.11811	0.07	0.75	1916.0	1921.7	1927.8 ± 1.3	0.6
B2	0.7	327	110.2	0.05	1.4	3552	0.016	0.346222	0.08	5.6360	0.12	0.11806	0.08	0.71	1916.5	1921.6	1927.1 ± 1.5	0.5
B3	1.2	148	49.5	0.05	1.1	3660	0.014	0.343269	0.07	5.5857	0.10	0.11802	0.06	0.76	1902.4	1913.9	1926.4 ± 1.2	1.2
B4	1.1	106	35.1	0.04	2.4	1072	0.011	0.342233	0.23	5.5713	0.25	0.11807	0.08	0.94	1897.4	1911.7	1927.2 ± 1.4	1.5
JP10z																		
B1	2.8	184	61.9	0.05	1.1	10692	0.016	0.345325	0.07	5.6231	0.09	0.11810	0.06	0.73	1912.2	1919.6	1927.7 ± 1.1	0.8
B2	3.0	257	86.5	0.04	0.5	34134	0.011	0.346582	0.07	5.6441	0.09	0.11811	0.05	0.80	1918.3	1922.9	1927.8 ± 0.9	0.5
B4	3.3	389	130.2	0.02	1.4	20261	0.007	0.346585	0.05	5.6433	0.07	0.11809	0.04	0.76	1918.3	1922.7	1927.6 ± 0.8	0.5
JP11z																		
B1	1.0	320	106.0	0.02	2.9	2418	0.007	0.342462	0.10	5.5780	0.13	0.11813	0.07	0.83	1898.5	1912.7	1928.1 ± 1.3	1.5
B3	0.7	645	217.4	0.03	1.6	6425	0.009	0.347918	0.07	5.6626	0.10	0.11804	0.07	0.70	1924.6	1925.7	1926.8 ± 1.3	0.1
B4	0.9	212	71.3	0.05	1.9	2270	0.014	0.344809	0.12	5.6123	0.14	0.11805	0.07	0.86	1909.8	1918.0	1926.9 ± 1.3	0.9

(a) B1, B2, etc. correspond to labels on concordia diagrams; all fractions are composed of single baddeleyite grains.

(b) Sample weights are estimated to within 40% using measured grain dimensions and density.

(c) Model Th/U ratio calculated from radiogenic $^{208}Pb/^{206}Pb$ ratio and $^{207}Pb/^{206}Pb$ age.

(d) Total weight of common Pb.

(e) Measured ratio corrected for spike and fractionation only. Mass fractionation correction of 0.20%/amu ± 0.04%/amu (atomic mass unit) was applied to single-collector Daly analyses (based on daily analysis of NBS-981) and a correction of 0.12%/amu ± 0.04% was applied to dynamic Faraday-Daly analyses.

(f) Corrected for fractionation, spike, blank and initial common Pb. Total procedural blank was typically 1-2pg for Pb and <0.1pg for U. Measured laboratory blank composition: $^{206}Pb/^{204}Pb$ = 19.10 ± 0.1, $^{207}Pb/^{204}Pb$ = 15.71 ± 0.1, $^{208}Pb/^{204}Pb$ = 38.65 ± 0.1 (2 sigma); initial common Pb composition was taken from the model of Stacey and Kramers (1975) and the interpreted age of the sample.

(g) Errors are 2 sigma, propagated using the algorithms of Ludwig (1980).

(h) Age calculations are based on the decay constants of Steiger and Jäger (1977); error in the $^{207}Pb/^{206}Pb$ age reported at 2 sigma.

(i) % discordance = 100 - (100 x ($^{206}Pb/^{238}U$ age)/($^{207}Pb/^{206}Pb$ age)).

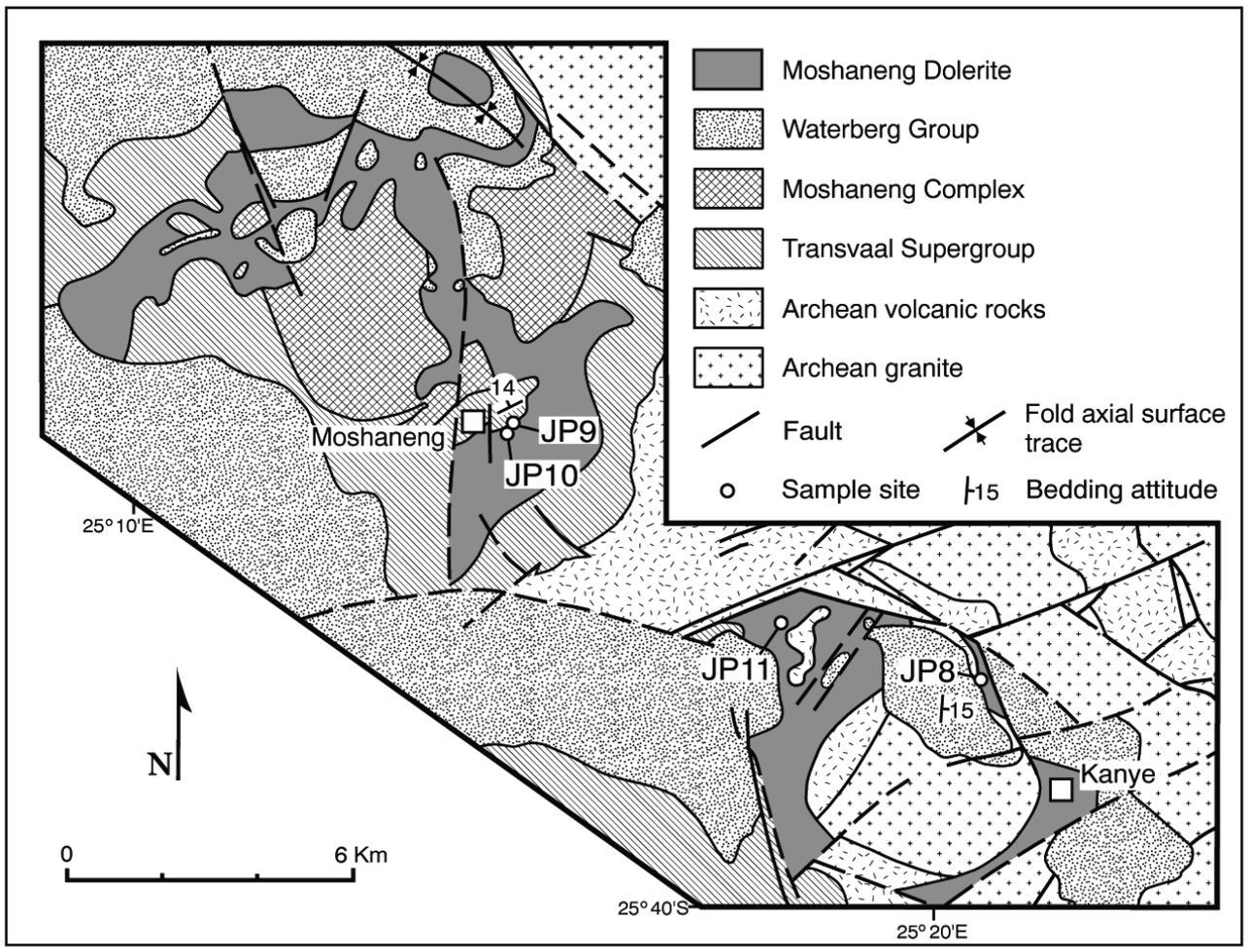


Figure 4. Geological map of the Moshaneng Dolerite in southeastern Botswana, from Aldiss *et al.* (1989). See Figure 1 for location. Sample localities are indicated.

hornblende and biotite. Variable degrees of low-temperature deuteric alteration are recorded by partial sericitization of plagioclase and partial replacement of mafic minerals by actinolite–epidote–chlorite–smectite intergrowths; rarely, plagioclase is replaced by prehnite. More differentiated rocks ranging from ferrodolerite to granophyre occur in the center or upper part of some intrusions. Irregular, centimeter- to decimeter-scale pegmatitic pockets within the dolerites contain curved blades of augite or pigeonite up to several centimeters in length intergrown with less common hornblende, abundant interstitial microgranophyre and discrete, fine-grained crystals of quartz and K-feldspar.

Geochronology

Baddeleyite was extracted from bulk-rock samples (up to 30kg) using standard crushing and mineral separation techniques. Isotopic analyses were carried out in the Isotope Geochemistry Laboratory at MIT. Single-crystal baddeleyite preparation, chemistry and analytical procedures follow those of Bowring *et al.* (1998). Additional analytical details and results are presented in Table 1 and Figure 5. All errors are reported at ± 2σ.

Three intrusions in the Waterberg Group in South Africa have yielded Paleoproterozoic ages. Sample WD04 was collected from dolerite containing small pegmatitic pockets and located within the upper portion of a sill intruding the Sterkrivier Formation west of Mokapane (formerly Potgietersrus) (Figure 2; du Plessis, 1978). Regression through five discordant analyses of baddeleyite from this sample (Figure 5a, Table 1) yields an upper intercept of 1873.7 ± 0.8 Ma (MSWD = 1.5), which we interpret as the igneous crystallization age, and a lower intercept of 201 ± 24 Ma.

Sample WD10 comes from an irregularly shaped sill up to 120m thick that intrudes the Setlaole, Makgabeng and Mogalakwena Formations farther to the north (Figure 2). The sill is composite and contains dolerite along the margins and more felsic rocks in the center (Strauss, 1947). A continuation of the same sill extends 28km farther west but is separated from the sample locality by a fault (Figure 2; Brandl, 1985; 1993). Sample WD10 was collected in the lower part of the sill, from dolerite containing abundant interstitial microgranophyre. Four baddeleyite crystals yielded discordant analyses (Figure 5b, Table 1), with a weighted mean $^{207}Pb/^{206}Pb$ date of 1878.8 ± 0.5 Ma (MSWD = 1.2), which we interpret as the crystallization age of the sill.

Sample WD27z comes from an irregular, composite body termed the Bakker's Pass intrusion by Jansen

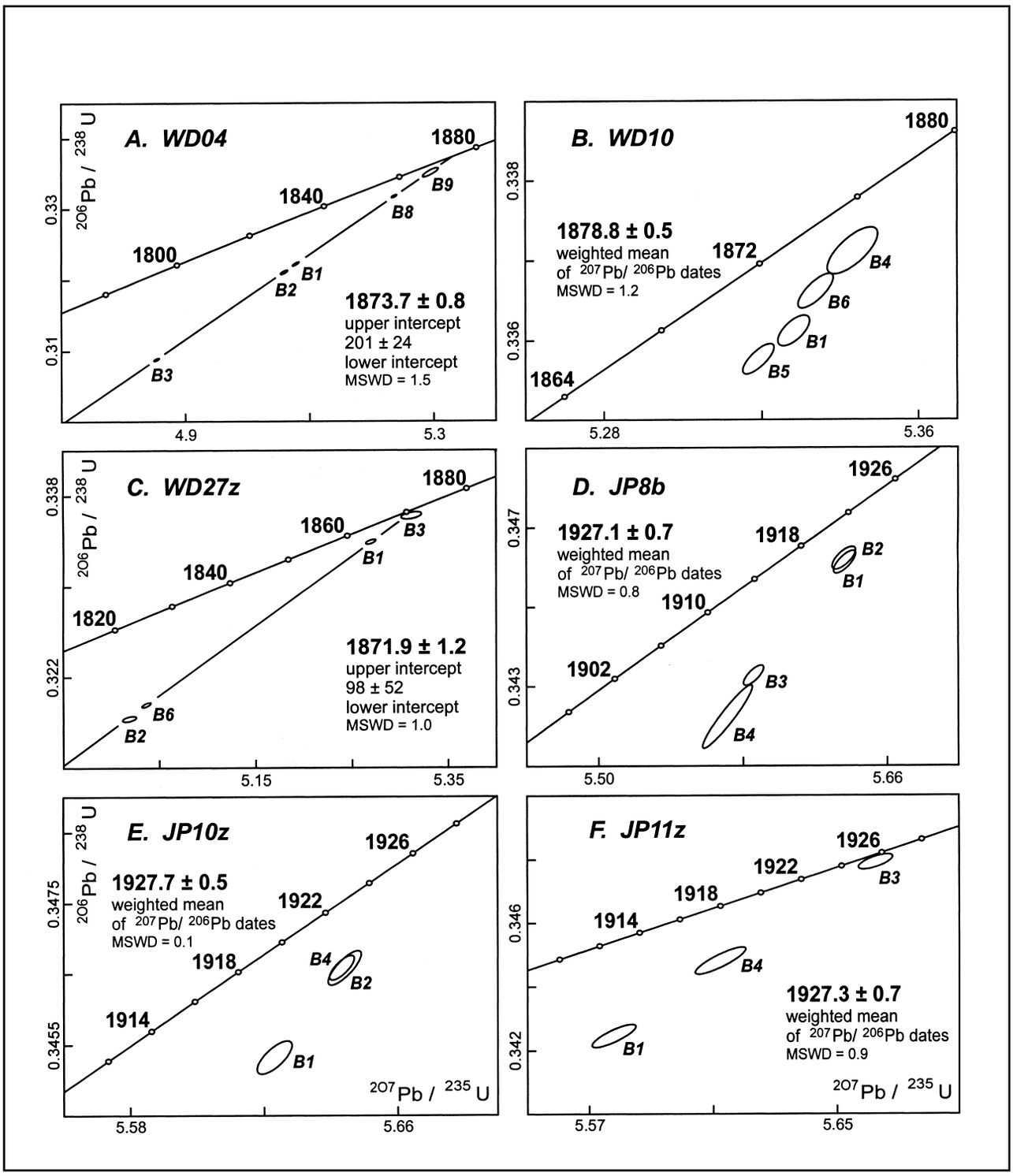


Figure 5. Concordia diagrams for samples of Paleoproterozoic dolerites intruding the Waterberg Group. Individual analyses are depicted as 2σ error ellipses. Numbers by analyses correspond to individual baddeleyite crystals analyzed (Table 1).

(1982), which cuts the Alma, Skilpadkop and Aasvoëlkop Formations northeast of Thabazimbi (Figure 2). Parts of the intrusion are discordant, but concordant upper and lower contacts are exposed along the road that follows Bakker's Pass down the Waterberg escarpment. The intrusion contains significant amounts of granophyre as well as dolerite and is inferred to connect beneath a short stretch of cover with a dolerite sill that extends for > 30km to the east (Figure 2; Jansen, 1982). Sample WD27z was collected from a pegmatitic pocket in the upper part of the intrusion along Bakker's Pass. Regression through four discordant baddeleyite analyses (Figure 5c, Table 1) yields an upper intercept of 1871.9 ± 1.2 Ma (MSWD = 1.0), which we interpret as the igneous crystallization age, and a lower intercept of 98 ± 52 Ma.

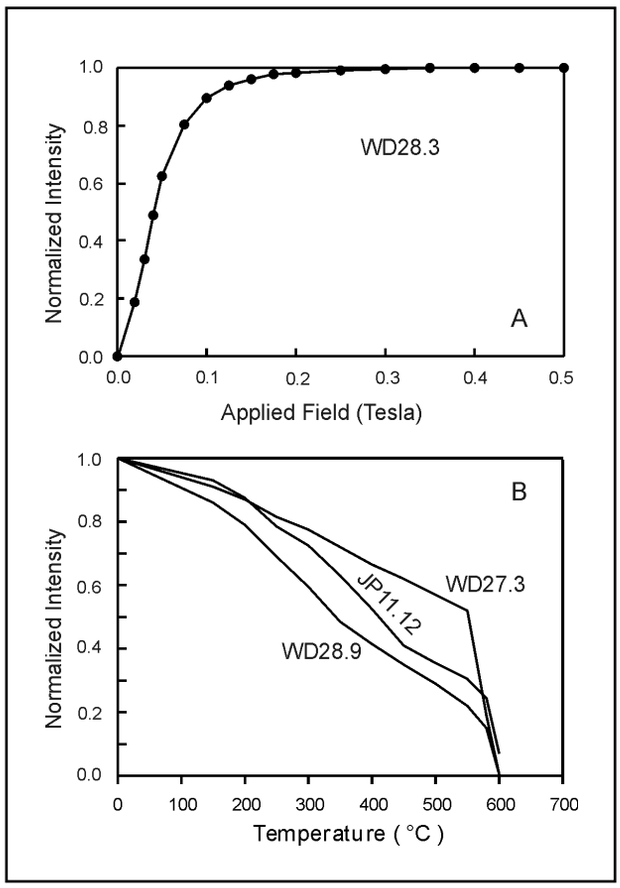


Figure 6. (**a**) Isothermal remanent magnetization (IRM) acquisition curve for representative WD sample. IRM acquisition curves identify magnetite as the dominant or sole magnetic mineral in the studied samples. (**b**) Thermal demagnetization of the natural remanent magnetization for representative WD and JP samples. All samples have unblocking temperatures typical for magnetite. In samples JP11.12 and WD28.9, a distinct second phase is apparent at intermediate temperatures.

In Botswana, samples were collected at three sites within the Moshaneng Dolerite, which comprises a series of faulted dolerite sheets and smaller intrusive bodies cutting the Waterberg Group and underlying units in the Kanye area (Figure 4). Aldiss *et al.* (1989) considered the various dolerite bodies to be parts of a single, large intrusion that was emplaced as the host rocks were faulted and folded to make space for the intruding magma. At site JP8, where the dolerite shows a chilled upper margin against the Mannyelanong Hill Formation, small pegmatitic pockets (sample JP8b) were collected from typical medium-grained dolerite in the inner part of the sill. Four baddeleyite crystals from the pegmatite yielded discordant analyses (Figure 5d, Table 1), with a weighted mean $^{207}Pb/^{206}Pb$ date of 1927.1 ± 0.7 Ma (MSWD = 0.8). Sample JP10z was collected within the Moshaneng Quarry, where the dolerite is currently being exploited for crushed stone. Three baddeleyite crystals yielded discordant analyses (Figure 5e, Table 1), with a weighted mean $^{207}Pb/^{206}Pb$ date of 1927.7 ± 0.5 Ma (MSWD = 0.1). Three baddeleyite crystals from sample JP11z also yielded discordant analyses (Figure 5f, Table 1), with a weighted mean $^{207}Pb/^{206}Pb$ date of 1927.3 ± 0.7 Ma (MSWD = 0.9). These data show that different parts of the Moshaneng Dolerite were emplaced in a narrow time interval at ~ 1927 Ma.

Paleomagnetic data

Paleomagnetic samples were collected at or near the same localities as the geochronological and geochemical samples. Previous experience with similar dolerites in southern Africa indicates that many of these rocks are very strongly magnetized, probably as a result of lightening strikes, and the paleomagnetic samples were therefore oriented with a sun compass and distributed over a large outcrop area. A minimum of nine samples were taken at each site using a gasoline-powered rock drill.

Paleomagnetic analyses were carried out at the University of Texas, Austin. IRM (isothermal remanent magnetization) acquisition experiments identify magnetite as the dominant or sole magnetic mineral in all samples (Figure 6a). This is confirmed by thermal demagnetization (Figure 6b), which indicates a broad range of unblocking temperatures below the magnetite Curie point of 580°C. In view of the directional data discussed below, it may be significant that data for sample JP11.12 suggest a second Curie temperature near 450°C, and data for sample WD28.9 suggest a Curie point near 350°C.

The samples were kept in a magnetically shielded room throughout the demagnetization procedure. All samples were subjected to progressive alternating field demagnetization up to 100mT, and the data were analyzed by the principal component method (Kirschvink, 1980). Most samples lose a significant amount of their intensity after demagnetization to 10mT. The directions of this component are random and thus represent a magnetization acquired after sampling. The characteristic directions are typically revealed after demagnetization to 40mT (Figure 7). Some samples exhibit well-defined linear trends towards the origin (*e.g.* WD13.8), whereas other samples do not change in intensity or direction in the available range of demagnetization fields (*e.g.* WD29.12). In the latter case, the Fisher mean of the high-coercivity directions was taken as the characteristic direction. In seven of the eleven samples from site WD28 (*e.g.* WD28.8 and WD28.11, Figure 7), a common direction of magnetization is observed over the demagnetization range from 20 to 60mT, which is distinctly different from the high-coercivity direction.

The site mean directions for the high-coercivity components of the WD sites from dolerite intrusions in South Africa that yielded usable paleomagnetic data are listed in Table 2 together with their statistical parameters. Samples from individual sites were excluded from the statistics if their direction deviated from the mean by more than two angular standard deviations or if no stable component could be identified. Site WD21 is

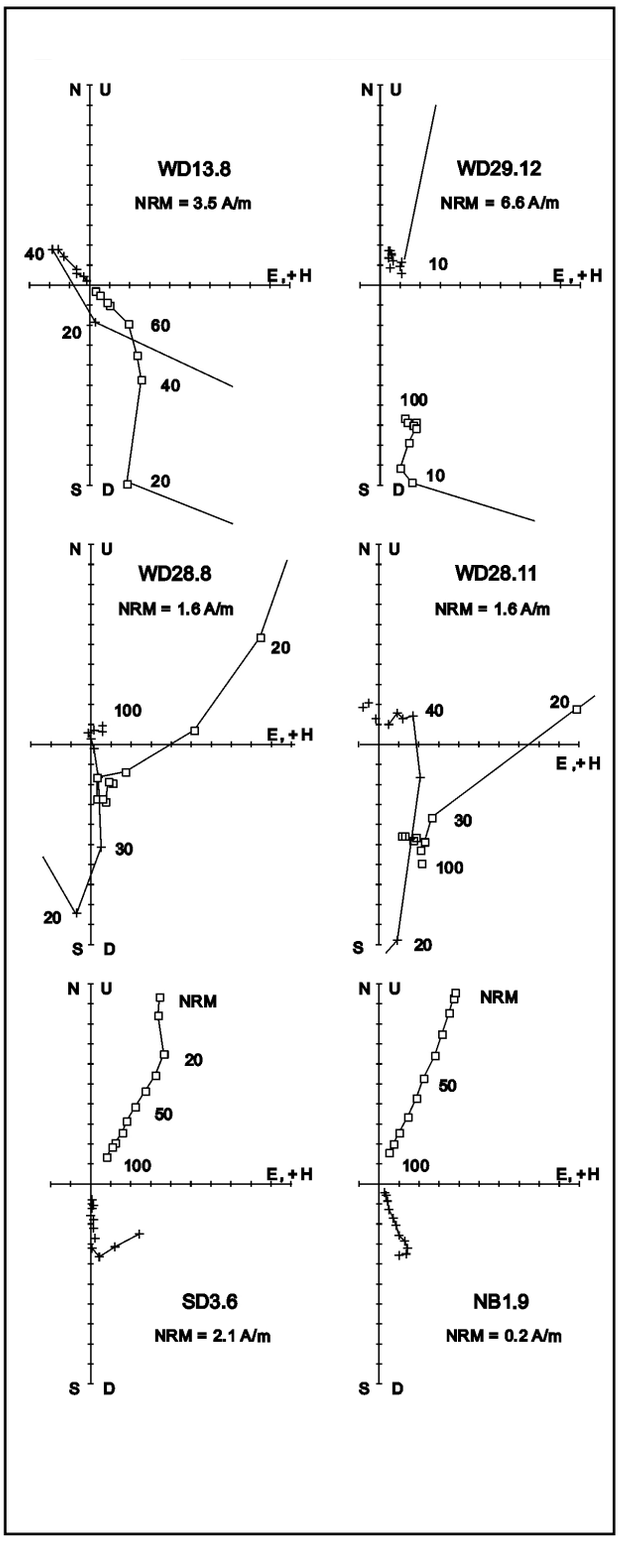


Figure 7. Vector component diagrams for representative WD samples from dolerite sills in the Waterberg Group in South Africa, and from a dolerite sill (SD3.6) and a basalt lava (NB1.9) in the Soutpansberg Group. Open symbols are the projection onto the Up-Down-Horizontal plane and crosses are in the North-South-East-West plane. Numbers indicate the demagnetization field in millitesla. The samples from site WD28 carry a distinct low-coercivity direction identified in the range from 20 to 60mT.

within a dolerite dyke ≤ 50m thick. Sites WD27 and WD28 were collected from different levels of the Bakker's Pass intrusion exposed in road cuts along the pass. The other three sites are within typical dolerite sills (Figure 2). The directions for sites WD21 and WD29 cluster tightly. Their large precision parameters (k) suggest that these sites did not average out secular variations. By contrast, comparison of k values between the different sites argues that the group mean does represent a time-averaged reading. Wherever possible, we measured the bedding attitude of nearby Waterberg strata and applied this tilt correction to the dolerites. Waterberg strata typically show low regional dips (≤ 10°) throughout the study area, indicating that structural effects are of negligible importance in interpreting the paleomagnetic data.

Basalt lavas and additional dolerite sills were collected in the Soutpansberg Group 80–230km east-northeast of the easternmost WD sites, in order to test whether the Soutpansberg mafic rocks (discussed further below) might have been emplaced during the same magmatic episode. The sampled dolerite sills intrude the Sibasa and Wyllies Poort Formations within the lower and middle parts of the group (Brandl, 1981; 1985). One lava sample (LB1, Table 2) is from the 3-km-thick Sibasa Basalt Formation near the base of the group. The other lava sample (NB1) is from a thinner volcanic unit within the stratigraphically higher Nzhelele Formation (Brandl, 1985). Alternating field demagnetization of samples from the two basalt flows and five dolerite sills (SD1–5) yielded very simple vector component diagrams (Figure 7) and well-defined means (Table 2). The site mean directions of these sites and the WD sites are depicted in Figure 8. The two sample sets are of opposite polarity. The F-test establishes that the two groups are indeed antipodal at the 95% significance level and thus can be combined to yield a single pole position, referred to here as pole WSD. The dual polarity of the data and their directional consistency over a distance of ~400km are good indications that the rocks carry a primary magnetization.

Sites JP9, JP10 and JP11 from the Moshaneng Dolerite in Botswana (Figure 4) yielded very different results. Most samples carry a well-defined characteristic direction of magnetization, typically revealed only at higher demagnetization fields (Figure 9). Sample JP10A.2 is a particularly striking example. The directions for site JP11 fall into two populations (Figure 10a); ten directions cluster in a different direction from sites JP9 and JP10, and ten scatter. This dichotomy is also apparent when the samples are sorted according to their NRM intensities (Figure 10b). With one exception, it is the high-intensity samples that scatter, and it seems likely that they were remagnetized by lightning. Most of the samples from other sites that were excluded from the statistics are also high-intensity samples. At sites JP9 and JP10, the dolerite intrudes Transvaal strata that were tilted prior to deposition of the Waterberg Group. Subsequent folding and faulting in the area are inferred

Table 2. Statistical parameters of paleomagnetic data

	Site	N	R	D	I	k	α95	Lat	Long	dp	dm
WD13	23.71S, 28.39E	8/1	7.674	333.3	53.9	21.5	12.2	26.8	3.9	12.0	17.1
WD21	23.19S, 28.73E	10/2	9.954	312.0	65.1	196.8	3.5				
		SC		309.3	59.3	196.8	3.5	11.0	351.6	3.9	5.2
WD23	23.42S, 28.80E	8/4	7.364	8.7	73.1	11.0	17.5	7.5	33.3	27.9	31.2
WD27	24.49S, 27.56E	11/1	10.252	319.4	74.8	13.4	13.0				
		SC	10.252	324.1	71.0	13.4	13.0	4.4	8.1	19.7	22.6
WD28	24.50S, 27.56E	11/1	10.396	336.1	75.8	16.6	11.6				
		SC	10.396	340.8	72.0	16.6	11.6	6.9	17.1	18.0	20.4
WD29	24.31S, 27.42E	12/0	11.928	28.7	79.9	152.3	3.5	353.2	36.8	6.5	6.8
SD1	22.59S, 30.87E	11/4	10.013	165.3	-57.4	10.1	15.1	27.8	17.8	16.1	22.0
SD2	22.60S, 30.88E	9/3	8.613	173.9	-60.8	20.7	11.6	25.3	25.9	13.5	17.7
SD3	22.73S, 30.18E	10/0	9.764	168.9	-68.0	38.2	7.9	15.5	23.0	11.2	13.3
SD4	23.02S, 29.60E	12/0	11.823	156.5	-54.0	62.3	5.5	28.3	7.7	5.4	7.8
SD5	23.06S, 29.59E	11/1	10.912	160.8	-49.6	113.1	4.3	33.5	9.7	3.8	5.7
LB1	22.95S, 30.46E	13/0	12.665	150.0	-57.0	35.9	7.0	23.2	4.9	7.4	10.2
NB1	22.56S, 30.87E	9/1	8.897	163.6	-69.3	77.5	5.9	13.1	20.8	8.6	10.0
Mean of Site Poles		13	12.474				8.9	15.6	17.1		
JP9	24.96S, 25.25E	14/0	13.677	324.1	7.2	40.2	6.3	45.0	329.4	3.2	6.4
JP10	24.90S, 25.25E	13/1	12.261	315.0	0.9	31.6	7.5	39.6	318.5	3.7	7.5
JP11	24.95S, 25.32E	10/10	9.870	180.5	-28.1	69.2	5.8	50.2	26.0	3.5	6.4
WD28 Low	24.50S, 27.56E	7/3	6.943	179.9	-42.3	104.4	5.9				
		SC	6.943	179.8	-38.3	104.4	5.9	44.0	27.3	4.2	7.0

N = number of samples used/rejected; R = resultant vector; D = declination; I = inclination; k = Fisher's precision parameter; α95 = 95% circle of confidence; Lat = north latitude; Long = east longitude; dp, dm = semi-axes of ellipse of confidence; SC = structurally corrected; W28 Low = low-coercivity component.

to have occurred during dolerite emplacement (Aldiss *et al.*, 1989), and we therefore have not corrected our data for tilt of nearby host strata.

The Botswana site mean directions are distinctly different from the WSD sites from South Africa (Figure 11, Table 2). The dolerite at site WD27 has a U–Pb crystallization age ~55m.y. younger than the dolerite at site JP10 (Figure 5), which could account for the different directions for those sites. The dolerite at site JP11 is identical in age to that at site JP10 (Figure 5), but its direction agrees neither with sites JP9 and JP10 nor with the WSD pole. Its direction is, however, statistically indistinguishable from the directions for the ~1.1 Ga Umkondo dolerites in Botswana and South Africa (Pancake, 2001; Gose *et al.*, 2003), which strongly suggests that site JP11 was remagnetized during the Umkondo igneous event. The low-coercivity component observed in samples from site WD28 has a similar direction and thus also likely records remagnetization during Umkondo activity. Disturbance of parts of the Moshaneng Dolerite in this time frame is suggested by a K–Ar date of 1110 ± 44 Ma (recalculated to currently recommended constants) for a composite plagioclase–pyroxene fraction from dolerite at the Moshaneng Quarry (Jones and McElhinny, 1966), near our sites JP9 and JP10.

The pole positions for our data are compared with relevant poles of similar age from southern Africa in Figure 12, and some of the poles shown there are discussed in the following section. Our WSD pole for the post-Waterberg dolerites in South Africa and the Soutpansberg mafic rocks falls between the pole for metamorphic rocks from the Limpopo Belt (LPA), the pole for the Sand River dykes (SR) in the same area (Morgan, 1985) and one of the poles (Seb-2) from the undated Sebanga dykes in Zimbabwe (Bates and Jones, 1996). Morgan (1985) inferred that Rb–Sr isotopic data from the Limpopo Belt record regional cooling at ~1.98 Ga and constrain the age of the Limpopo pole, but this needs to be tested by detailed thermochronological studies. The only available geochronological constraint for the Sand River dykes is a whole-rock Rb–Sr errorchron date of 1876 ± 68 Ma (Barton, 1979), but the geological significance of this result is unclear. Precise isotopic age constraints on the rock units that have yielded the paleomagnetic data are needed in order to develop a well-constrained apparent polar wander path for southern Africa in this time frame. Our WSD pole, dated at 1879–1872 Ma, provides a step in this direction.

Paleomagnetic data from the Bushveld Complex have been widely used in developing the Paleoproterozoic apparent polar wander path for southern Africa. In Figure 12, we show the poles derived from specific sampling areas within the complex (Hattingh, 1986a; b; c; 1989; Hattingh and Pauls, 1994). We also show the mean of these poles, because U–Pb geochronology indicates that the Bushveld Complex was emplaced in a narrow time frame at ~2.06 to ~2.05 Ga (Walraven and Hattingh, 1993; Walraven, 1997; de Waal

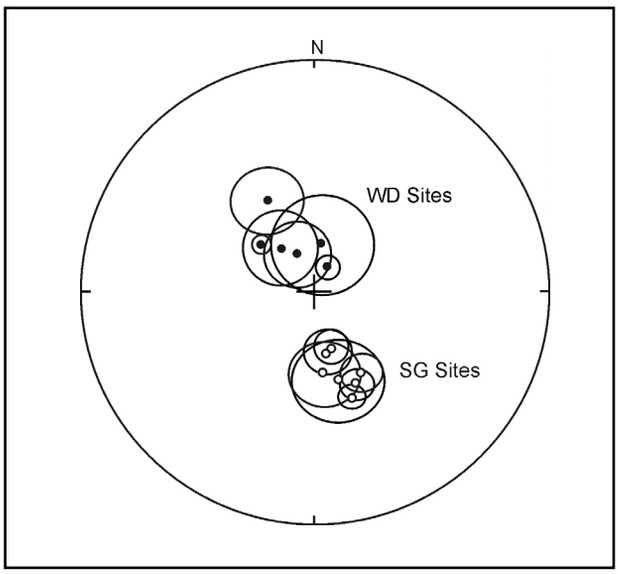


Figure 8. Site mean directions for post-Waterberg dolerites (WD) and Soutpansberg lavas and sills (SG), grouped together as WSD sites in the text, with their 95% circles of confidence. Lower hemisphere equal-area projection. Open (solid) symbols are in the upper (lower) hemisphere.

and Armstrong, 2000; Buick *et al.*, 2001; Eglington and Armstrong, 2004). Evans *et al.* (2002) averaged the individual Bushveld sampling sites, but that approach is biased by results from one area with a large number of sites. Before the Bushveld pole can be considered firmly established, it needs to be determined whether the scatter in pole positions indicates variations in the time at which the primary remanence was acquired during cooling (*cf.* Hattingh and Pauls, 1994), or whether the scatter is due to post-crystallization thermal disturbances and/or unresolved structural complexities.

The error circle of our WSD pole overlaps the error circle of the mean Bushveld pole. This suggests the possibility that parts of the Bushveld Complex were remagnetized due to widespread thermal disturbance or fluid migration occurring at the same time as emplacement of the ~1.88 to ~1.87 Ga post-Waterberg dolerite sills and coeval Soutpansberg mafic rocks. Further support for this suggestion comes from the fact that the Phalaborwa Complex (Figure 1), which has yielded U–Pb baddeleyite ages of 2059.6 ± 0.4 to 2060.6 ± 0.5 Ma (Reischmann, 1995; French *et al.*, 2002) and therefore is coeval with the Bushveld Complex, has a pole (PA1) that lies well to the northeast of the mean Bushveld pole (Figure 12).

The poles for sites JP9 and JP10 fall well outside the main trend. Jones and McElhinny (1966) reported pole positions for two nearby sites in the Moshaneng Dolerite (A1 and A2, Figure 12). One of their pole positions is similar to the JP9 and JP10 poles, but the other one is significantly different, which may reflect post-intrusion structural effects in the Moshaneng area. The fact that some of the Sebanga dykes from Zimbabwe yield a pole position (Bates and Jones, 1996) that is statistically

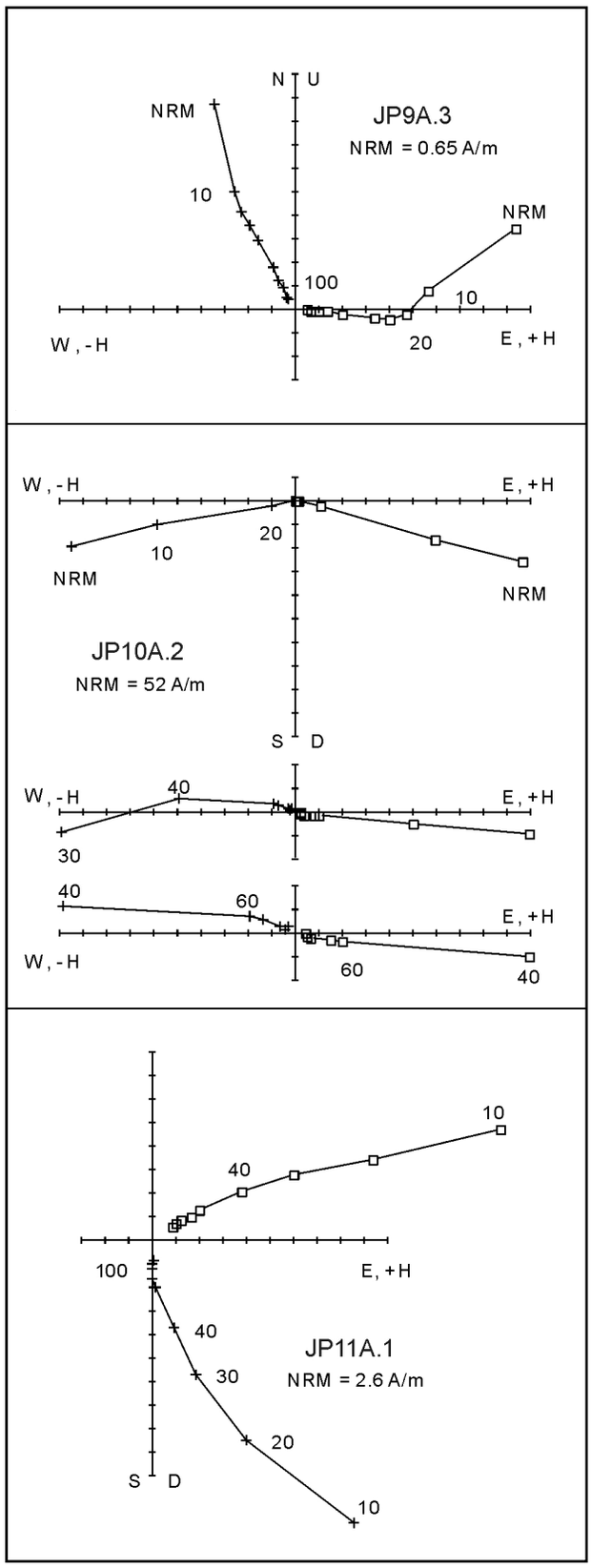


Figure 9. Vector component diagrams for representative JP samples from the Moshaneng Dolerite in Botswana. Open symbols are the projection onto the Up-Down-Horizontal plane and crosses are in the North-South-East-West plane. The results from sample JP10A.2 are displayed with two enlargements. Although the intensity has been reduced to 1/1000th of its NRM value, the characteristic direction is revealed only above 60mT.

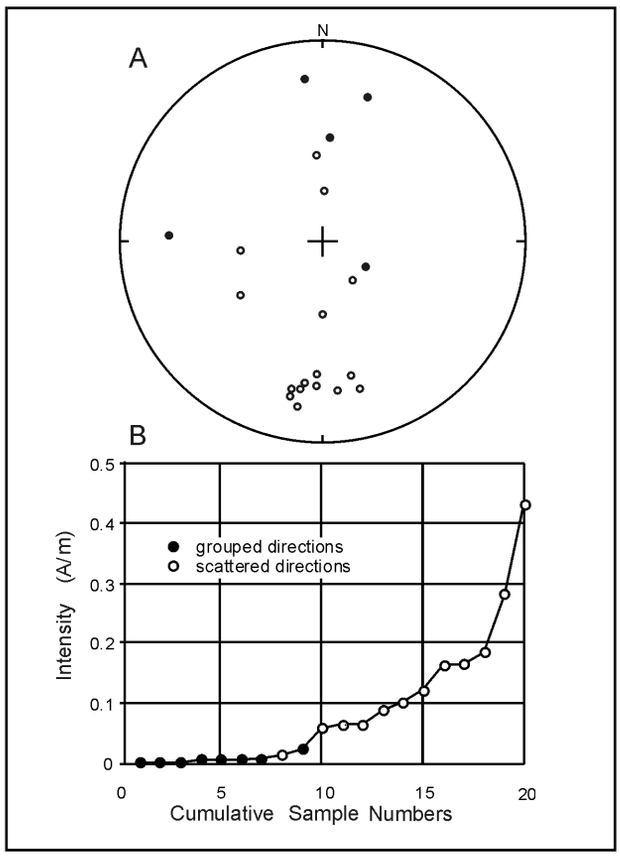


Figure 10. (**a**) Equal-area projection of the directions of magnetization for site JP11. Open (solid) symbols are in the upper (lower) hemisphere. (**b**) Samples sorted by their NRM intensities. All but one of the stronger samples had scattered directions, suggesting that these samples were remagnetized by lightning. Note that filled circles in *b* do not correspond to those in *a*.

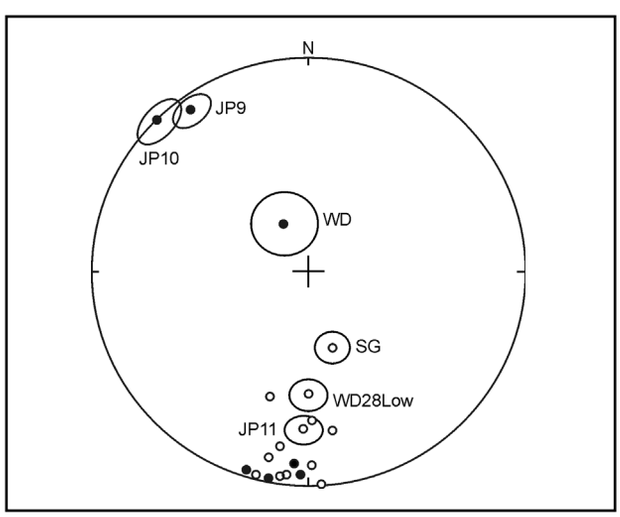


Figure 11. Site mean directions with their 95% circles of confidence for JP sites in Botswana and the low-coercivity component of site WD28 in South Africa. WD and SG are the site means of the post-Waterberg dolerites and Soutpansberg sites in South Africa. Individual points without error circles are site mean directions for Umkondo dolerites from Botswana and South Africa (Pancake, 2001; Gose, unpublished data). Open (solid) symbols are in the upper (lower) hemisphere.

indistinguishable from the JP9, JP10 and A1 poles makes it highly unlikely that these three directions from the Moshaneng Dolerite are the result of local structural effects. It is not obvious how to integrate these poles with the other poles in Figure 12 into a common apparent polar wander path.

Geochemistry

Preliminary results of geochemical studies by Bullen (in preparation) are presented here in order to characterize the nature of the mafic magmatism represented by our dated samples, and to allow comparison with other, possibly related Paleoproterozoic mafic suites in the region. The ~1.88 to ~1.87 Ga post-Waterberg dolerites in South Africa and the ~1.93 Ga Moshaneng Dolerite in Botswana are sub-alkali basaltic to rarely andesitic in composition (Table 3), with MgO values of ~9 to ~4.5 weight % (Figure 13), and with a clear Fe-enrichment trend. They range from olivine- to quartz-normative, with abundant normative hypersthene (based on $Fe^{2+}/(Fe^{2+} + Fe^{3+}) = 0.85$). Major-element contents are typical of continental tholeiites (*cf.* Irvine and Baragar, 1971; Le Maitre, 1989), and ratios of Ti/Zr (~60; Figure 14a) and Nb/Y (~0.25) (*cf.* Winchester and Floyd, 1977) indicate tholeiitic affinities. REE patterns are also typical of continental tholeiites and show moderate LREE enrichment (Figure 15a, b).

The ~1.93 and ~1.88 to ~1.87 Ga dolerites are very similar in petrography, degree of deuteric alteration and major-element chemistry to the ~1.1 Ga Umkondo dolerite intrusions present in the same region (Table 3; Bullen, unpublished data). However, differences in incompatible element geochemistry discriminate between the different age groups and may be useful in future attempts to delineate their respective regional extents. For example, the ~1.1 Ga dolerites have lower Ti/Zr, Ti/Hf and Ti/La ratios than their older counterparts (Figure 14), as well as different REE profiles (Figure 15c). Although all of the samples have similar La_N/Lu_N values, the ~1.1 Ga dolerites typically are relatively enriched in REEs compared to the ~1.88 to ~1.87 Ga dolerites. Because the MgO range of all the analysed samples is similar, the difference in REE abundances is not attributable to varying degrees of differentiation and suggests that the ~1.88 to ~1.87 Ga dolerites were derived from higher percent partial melting of a similar mantle source to that which produced the ~1.1 Ga dolerites. The ~1.1 Ga dolerites also display a marked negative Eu anomaly, which is absent in the older dolerites and, together with the relatively low Sr abundances (Figure 14b), is inferred to record plagioclase fractionation prior to intrusion.

Discussion

Age and regional relations of the Waterberg Group

The U–Pb baddeleyite age of ~1927 Ma reported here for the Moshaneng Dolerite provides a younger age limit for the Mannyelanong Formation in the lower part

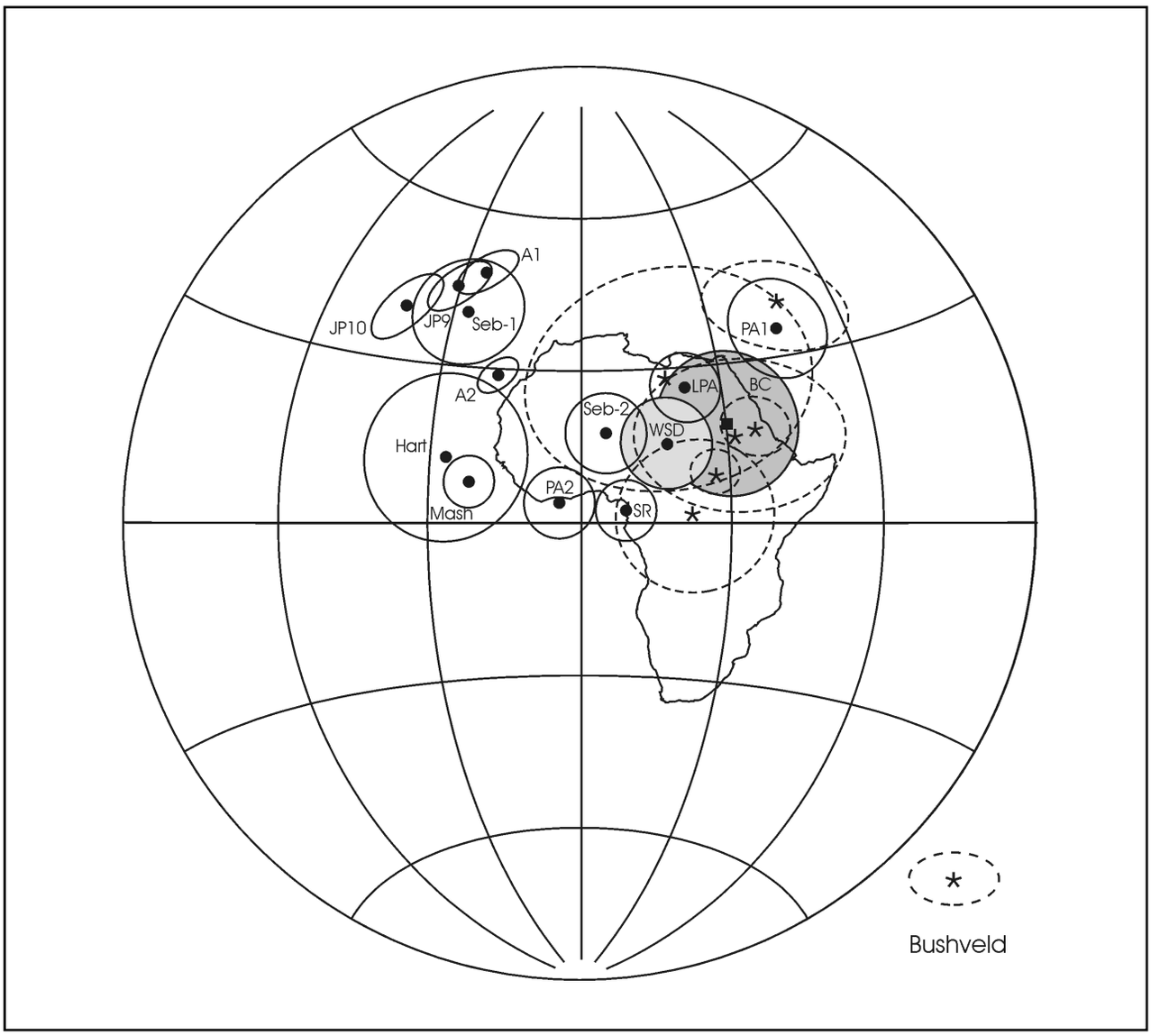


Figure 12. Pole positions with their 95% error limits for our combined WSD sites (shaded light grey) and two JP sites, together with published poles. The poles from different parts of the Bushveld Complex (star symbols) are combined to yield a mean Bushveld pole (BC, square symbol), with its error circle shaded dark grey. A1, A2: Moshaneng Dolerite (Jones and McElhinny, 1966); Hart: Hartley Basalt Formation (Evans *et al*., 2002); LPA: Limpopo gneiss group A (Morgan, 1985); Mash: Mashonaland dolerites (Bates and Jones, 1996); PA1, PA2: Phalaborwa Complex (Morgan and Briden, 1981); Seb-1, Seb-2: Sebanga dykes in Zimbabwe (Bates and Jones, 1996); SR: Sand River dykes (Morgan, 1985). See text for age controls on poles. Pole PA1 is considered to record the primary remanence acquired during emplacement of the main phases of the Phalaborwa Complex. Pole PA2 comes mainly from dolerite dykes that intrude the complex (see discussion in Evans *et al*., 2002).

of the Waterberg Group in southeastern Botswana. The U–Pb age of ~1879 Ma for dolerite sample WD10 in South Africa (Figure 2) constrains the Mogalakwena Formation and underlying Waterberg units to be pre-1879 Ma. As discussed earlier, the Mannyelanong Formation is correlated with the Skilpadkop Formation (and laterally equivalent Setlaole Formation) in South Africa (Figure 3). Whether the ~1927 Ma minimum age for the Mannyelanong Formation also applies to the equivalent Waterberg succession in South Africa depends on the validity of these lithostratigraphic correlations, and the extent to which the units are time-transgressive.

Based partly on the local abundance of volcanic rocks, several authors have proposed that the oldest parts of the Waterberg Group in South Africa (*e.g.* Swaershoek and lower Sterkrivier Formations) were deposited during waning stages of volcanism linked to the Bushveld magmatic event at ~2.06 Ga (*e.g.* Coertze *et al*., 1977; Cheney and Twist, 1986; Callaghan *et al*., 1991). This hypothesis is consistent with the new age constraints presented herein for the Waterberg Group but needs confirmation by geochronological studies on the Waterberg volcanic rocks.

Additional age constraints are provided by relations between the northern part of the Waterberg Group and

Table 3. Geochemical analyses of representative samples of post-Waterberg dolerites

	1.88–1.87 Ga dolerite sills				Moshaneng dolerite			1.1 Ga Umkondo sills		
Sample	**DBO1.43**	**DBO1.64**	**DBO1.72**	**DBO1.115**	**JP8**	**JP9**	**DBO1.4**	**DBO1.6**	**DBO1.19**	**DBO1.52**
SiO_2	51.06	51.79	50.75	51.06	51.76	50.13	53.40	50.94	54.72	53.86
TiO_2	0.78	1.00	0.72	0.79	1.25	1.59	0.89	0.76	0.95	0.79
Al_2O_3	14.52	15.86	15.02	16.02	15.75	14.89	14.54	15.16	14.35	14.66
Fe_2O_3	10.25	11.38	10.26	10.10	15.00	14.98	10.89	10.46	11.79	10.44
MnO	0.16	0.16	0.15	0.14	0.20	0.18	0.16	0.17	0.17	0.17
MgO	6.95	5.26	7.47	5.65	5.55	5.33	5.44	7.12	5.03	6.31
CaO	11.75	9.98	11.43	11.18	9.55	9.63	9.31	10.04	9.17	8.64
Na_2O	1.95	2.29	1.94	2.11	0.88	0.94	1.98	1.98	2.14	1.89
K_2O	0.72	0.82	0.57	0.77	0.81	0.89	1.31	0.93	1.43	1.12
P_2O_5	0.07	0.08	0.06	0.04	0.14	0.19	0.11	0.08	0.12	0.09
LOI	0.79	1.34	0.95	1.21	0.65	0.91	1.39	1.33	0.39	2.07
Total	99.46	99.96	99.32	99.07	101.54	99.66	99.42	98.97	100.26	100.04
Sc	29	24	26	25	32	24	35	40	35	29
V	237	237	203	193	320	336	247	222	254	225
Cr	126	17	198	100	22	151	62	154	59	166
Ni	112	84	164	266	81	115	77	116	92	68
Zn	68	88	74	71	103	113	73	95	90	110
Rb	27	35	17	23	48	37	58	41	66	42
Sr	185	206	187	205	152	160	145	139	126	206
Y	17	21	16	16	32	35	29	24	30	26
Zr	68	92	68	74	108	147	126	104	137	120
Ba	158	146	135	164	177	215	291	na	na	407
Nb	3.29	4.34	3.09	3.44	3.70	5.73	5.30	4.31	5.41	5.00
La	8.87	9.31	8.13	8.27	12.00	15.76	17.32	13.71	18.61	15.71
Ce	19.12	21.53	17.51	17.99	26.07	34.89	38.26	29.84	41.38	34.42
Pr	2.36	2.56	2.22	2.30	3.44	4.35	4.75	3.66	5.05	4.01
Nd	10.66	11.64	9.80	10.22	15.35	19.48	19.79	15.36	21.23	16.65
Sm	2.55	2.84	2.33	2.42	4.01	4.78	4.40	3.48	4.86	3.59
Eu	0.87	0.90	0.79	0.80	1.24	1.54	1.14	0.91	1.20	0.97
Gd	2.75	2.87	2.60	2.57	4.32	5.32	4.43	3.50	4.96	3.58
Tb	0.47	0.47	0.45	0.44	0.74	0.90	0.71	0.56	0.86	0.60
Dy	2.81	2.89	2.67	2.64	4.47	5.45	4.45	3.52	5.28	3.61
Ho	0.57	0.56	0.51	0.52	0.95	1.12	0.95	0.76	1.14	0.78
Er	1.48	1.56	1.47	1.39	2.75	3.03	2.82	2.24	3.29	2.13
Tm	0.23	0.23	0.20	0.20	0.37	0.44	0.39	0.30	0.45	0.34
Yb	1.41	1.50	1.34	1.33	2.46	2.94	4.27	2.01	3.02	2.35
Lu	0.22	0.24	0.21	0.20	0.39	0.45	0.41	0.33	0.49	0.35
Hf	1.93	1.98	1.76	1.86	2.85	3.81	3.32	2.58	3.69	2.95
Th	2.43	2.71	2.21	2.03	2.73	3.11	5.57	4.34	6.34	4.77
U	0.63	0.70	0.53	0.61	0.62	0.77	0.95	0.73	1.13	0.86

Major elements in wt % oxide, trace elements in ppm. Total Fe as Fe_2O_3; na = not analyzed.

Major and some trace elements for Moshaneng Dolerite from Ward (2002).

Major and trace elements Sc through Ba analyzed by XRF at the School of Earth & Environmental Sciences, University of Portsmouth; trace elements Nb through U analyzed by ICP-MS at the Southampton Oceanography Centre, University of Southampton.

the Central Zone of the Limpopo Belt, which records both Archean and Paleoproterozoic high-grade metamorphism and ductile deformation. A variety of geochronometers document plutonism, anatexis and high-grade metamorphism at ~2.02 to ~2.0 Ga in the Central Zone, with cooling after the metamorphic peak extending to ≤ 1.8 Ga (*e.g.* Barton and van Reenen, 1992; Kamber *et al.*, 1995; Jaeckel *et al.*, 1997; Holzer *et al.*, 1998; 1999; McCourt and Armstrong, 1998; Kröner *et al.*, 1999; Schaller *et al.*, 1999; Chavagnac *et al.*, 2001; Buick *et al.*, 2003; Eglington and Armstrong, 2004). The southern boundary of the Central Zone is defined by the Palala Shear Zone, which underwent Paleoproterozoic ductile deformation followed by brittle reactivation to form the Melinda and related faults (Figure 2; Brandl, 2001; Bumby *et al.*, 2001a; 2002). The Palala Shear Zone trends beneath Waterberg strata northeast of Ellisras (Figure 2; Jansen, 1982; Brandl,

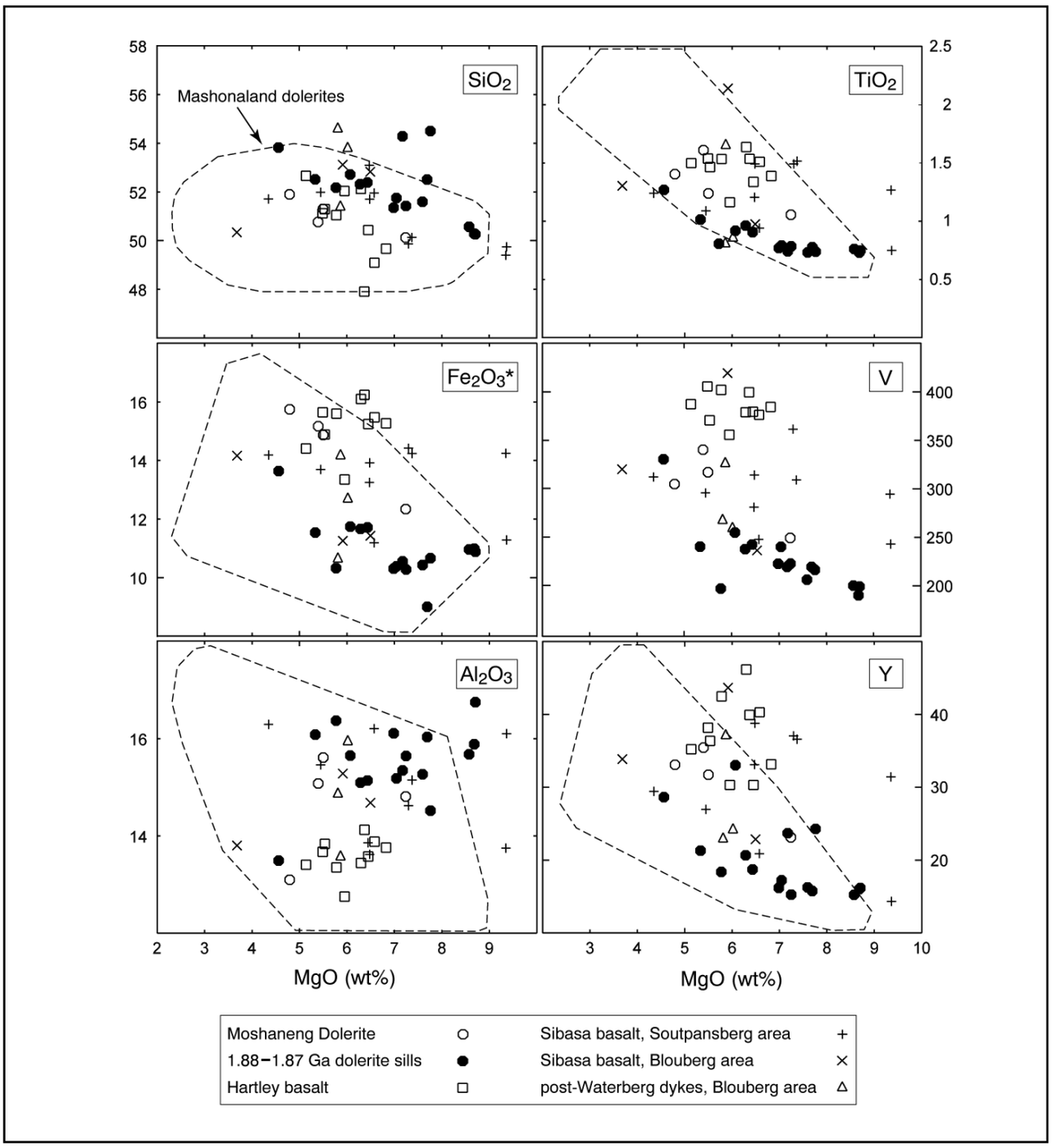


Figure 13. Selected major and trace elements plotted versus MgO for the Moshaneng Dolerite, ~1.88 to ~1.87 Ga post-Waterberg dolerite sills (Ward, 2002; Bullen, unpublished data), Hartley basalts (Cornell *et al.*, 1998), Sibasa basalts from Soutpansberg and Blouberg areas (Crow and Condie, 1990; Bumby *et al.*, 2001b) and dykes in the Blouberg area (Bumby *et al.*, 2001b). Range of data for Mashonaland dolerites (Stubbs *et al.*, 1999) is shown by outlined fields. Major oxides in weight %, trace elements in ppm.

1985). Farther east, in the Blouberg area, gneisses within the shear zone are unconformably overlain by the Blouberg Formation (Figure 2), which is inferred to have accumulated in a pull-apart basin during brittle faulting along the shear zone (Bumby *et al.*, 2001a). Deformed Blouberg strata are unconformably overlain by gently dipping strata assigned to the Mogalakwena Formation of the Waterberg Group by Bumby *et al.* (2001a; b; 2002).

In the literature, isotopic dates from the deformed Palala Granite within the Palala Shear Zone have commonly been used to constrain the maximum age for unconformably overlying Waterberg strata (*e.g.* Jansen, 1982; Cheney and Twist, 1986). Burger and Coertze (1975, cited in Walraven *et al.*, 1990) reported a U–Pb zircon date of 1770 ± 30 Ma for the Palala Granite, but more recent work indicates that the granite is part of the ~2.06 Ga Bushveld Complex (Walraven *et al.*, 1990;

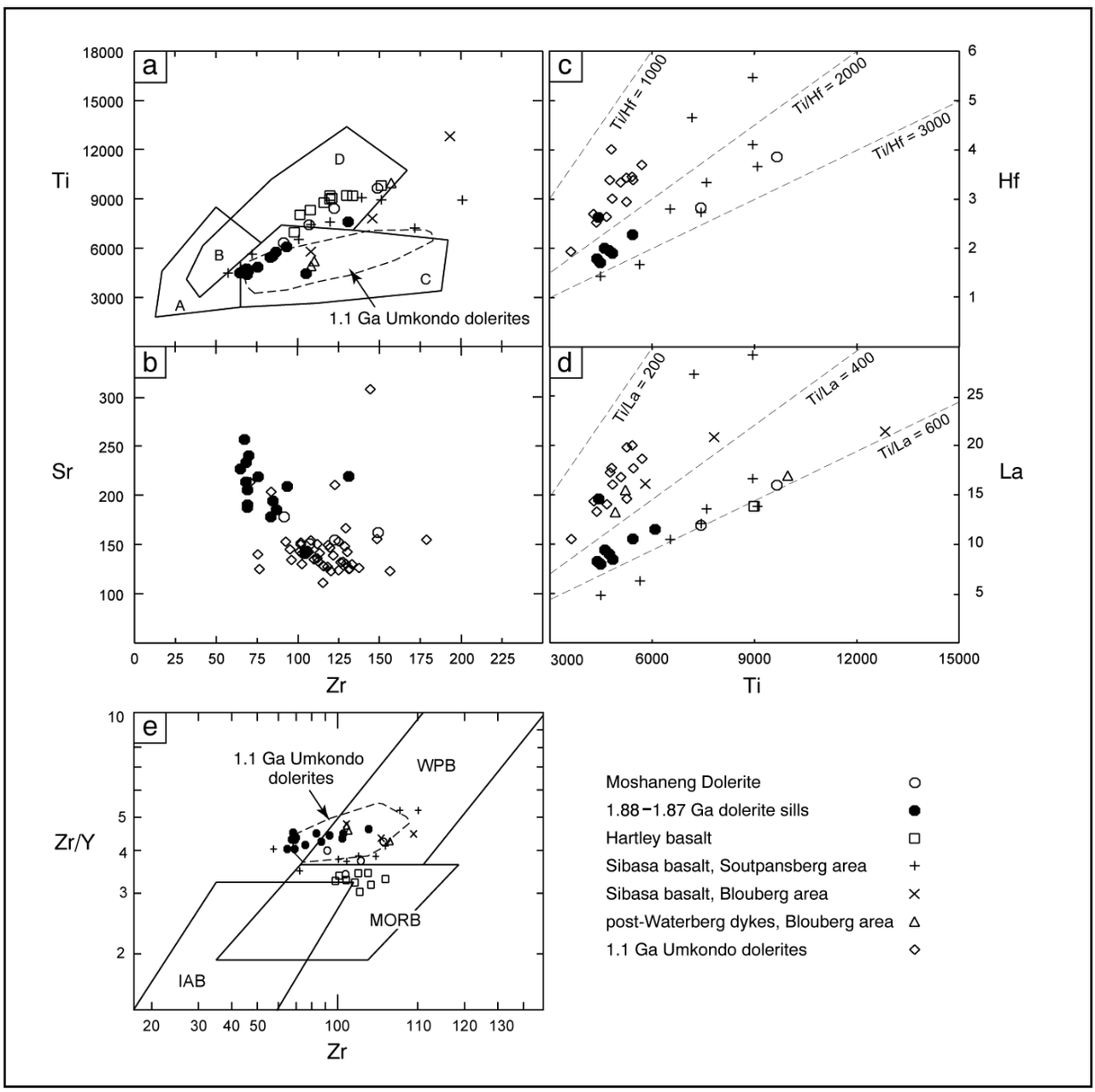


Figure 14. Incompatible-element variation diagrams for the Moshaneng Dolerite, ~1.88 to ~1.87 Ga post-Waterberg dolerites and 1.1 Ga Umkondo dolerites present in the same region (Ward, 2002; Bullen, unpublished data). Data for ~1.1 Ga Umkondo dolerites are indicated by outlined areas in *a* and *e*. Data for dykes in the Blouberg area (Bumby *et al.*, 2001b) and Sibasa and Hartley basalts (Crow and Condie, 1990; Cornell *et al.*, 1998; Bumby *et al.*, 2001b) are also shown. Fields in figure *a* (Pearce and Cann, 1973): A and B: low-potassium tholeiites; C and B: calc-alkali basalts; D and B: ocean-floor basalts. Fields in *e* (Pearce and Norry, 1979): WPB: within-plate basalts; IAB: island-arc basalts; MORB: mid-ocean-ridge basalts. No direct tectonic significance is attached to where individual samples plot in these two figures, because continental tholeiites typically spread across various fields on these discrimination diagrams (*e.g.* Wang and Glover, 1992).

Brandl, 2001). Schaller *et al.* (1999) reported a Rb–Sr biotite date of 1971 ± 26 Ma from greenschist-facies mylonites within the Palala Shear Zone. Additional constraints on the duration of ductile deformation in the shear zone are needed, but the available evidence, including our U–Pb crystallization age of ~1879 Ma for the dolerite sill (sample WD10) that intrudes the Setlaole, Makgabeng and Mogalakwena Formations, suggests that deposition in the northern part of the main Waterberg basin occurred between ~1.97 and ~1.88 Ga. If the minimum age of ~1927 Ma for the Mannyelanong Formation in Botswana can be applied to the correlative Skilpadkop and Setlaole Formations in South Africa, then this age bracket could be tightened. However, given the inherent problems in long-distance litho-stratigraphic correlation in fluvial continental successions, such regional extrapolations of the limited geochronological constraints are unwarranted at this stage.

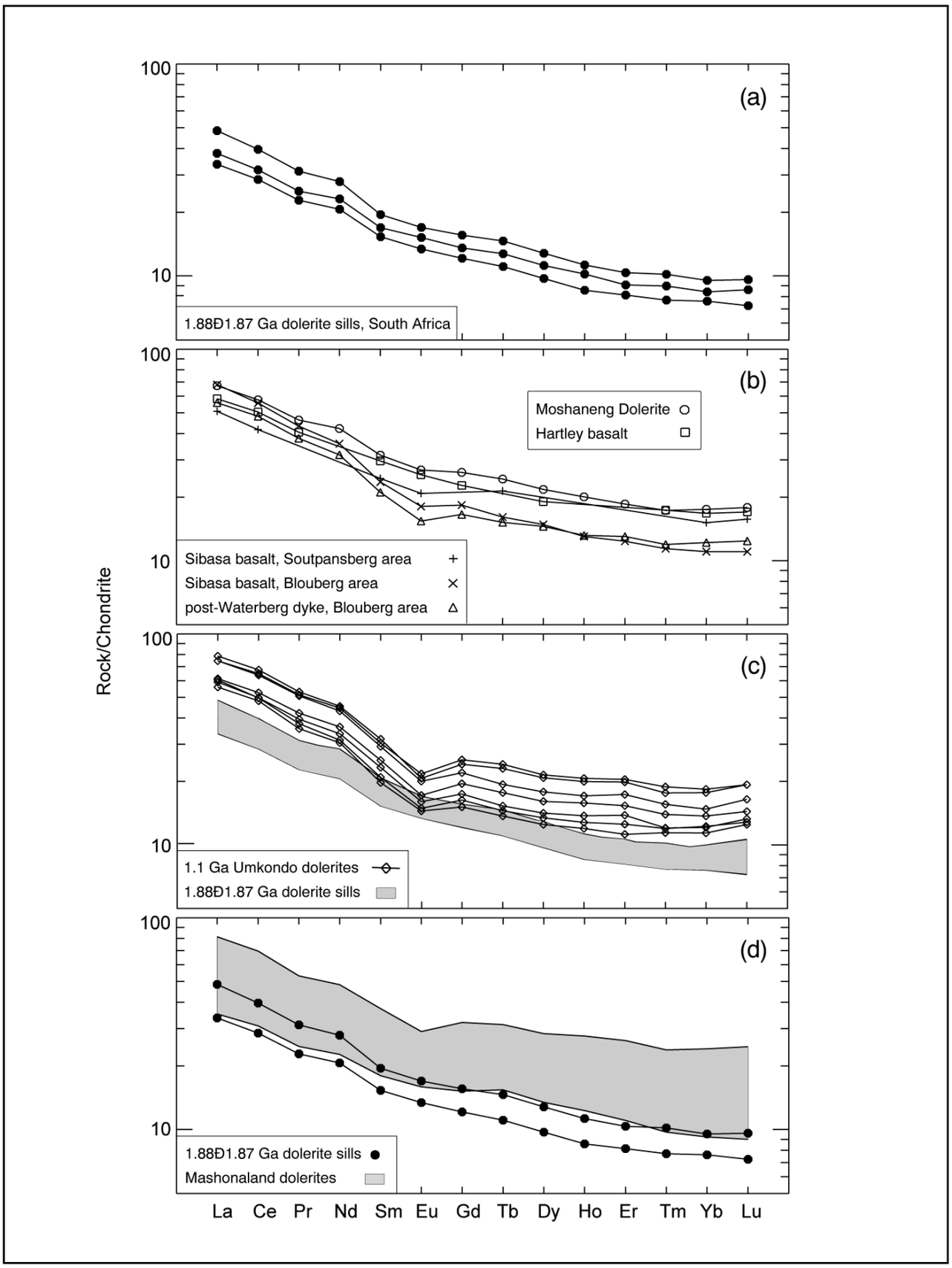


Figure 15. Chondrite-normalised rare-earth element distribution patterns for representative samples of (**a**) ~1.88 to ~1.87 Ga post-Waterberg dolerite sills, (**b**) Moshaneng Dolerite (Bullen, unpublished data), Hartley and Sibasa basalts (Crow and Condie, 1990; Cornell *et al.*, 1998; Bumby *et al.*, 2001b) and a dyke intruding the northern Waterberg Group (Bumby *et al.*, 2001b), (**c**) ~1.1 Ga Umkondo dolerites in Botswana and South Africa (Ward, 2002; Bullen, unpublished data), (**d**) Mashonaland dolerites (Stubbs *et al.*, 1999). Normalisation values from Sun and McDonough (1989).

These age constraints support previous interpretations (*e.g.* Jansen, 1982; Callaghan *et al.*, 1991; Callaghan, 1993; Bumby *et al.*, 2001a; b; 2002) that accumulation of significant parts of the Waterberg Group overlapped in time with tectonothermal activity and subsequent cooling at ~2.0 to ~1.8 Ga in the Limpopo Belt, with exhumation in the Central Zone of the belt supplying large volumes of detritus to the Waterberg basin. Other parts of the group accumulated during active faulting along the southern basin margins, at least partly in a transcurrent regime (Jansen, 1982; du Plessis, 1987; Callaghan *et al.*, 1991). This intracratonic tectonism records reactivation of older lines of structural weakness, including the Thabazimbi–Murchison Lineament, and was probably linked to coeval deformation in the Limpopo Belt.

Cheney and Twist (1986) divided the Waterberg Group into five unconformity-bounded sequences (Figure 3), the implication being that strata of significantly different ages may be present. Our geochronological data indicate that the Mogalakwena Formation and underlying units in the Waterberg Group were deposited prior to ~1879 Ma, but the depositional ages of the Cleremont and Vaalwater Formations at the top of the succession are not directly constrained because they are not in contact with one of the dated sills. However, the boundaries between these two units, and between the Cleremont Formation and underlying Mogalakwena Formation, are described as conformable and gradational (Jansen, 1982; Callaghan *et al.*, 1991), and Cheney and Twist (1986) included the two uppermost Waterberg formations in the same sequence as the Mogalakwena Formation (Figure 3). Hence, the younger age limit of ~1879 Ma for the Mogalakwena Formation probably applies to the entire Waterberg Group.

Stratigraphic relations between the Waterberg Group and other post-Bushveld Proterozoic successions shown in Figure 1 have generally been ambiguous because of the lack of good constraints on depositional ages. Jansen (1976) inferred that the middle and upper parts of the Soutpansberg Group are younger than the Waterberg Group, whereas other workers (*e.g.* Meinster, 1977; Brandl, 1987; Cheney *et al.*, 1990) proposed a pre-Waterberg age for the Soutpansberg Group. Based on detailed mapping in the Blouberg area, Bumby *et al.* (2001a, b, 2002) have recently argued that the Waterberg Group is unconformably overlain by the Soutpansberg Group. Their interpretation is consistent with the paleomagnetic data presented herein, which show that 1879–1872 Ma dolerites were intruded into Waterberg strata in the same time frame that basalts in the lower and middle parts of the Soutpansberg Group were being extruded.

Numerous workers (*e.g.* Jansen, 1981; Tankard *et al.*, 1982; Callaghan *et al.*, 1991; Cornell *et al.*, 1998) have suggested correlations between the Waterberg Group and the Olifantshoek Supergroup, which comprises a succession of dominantly clastic rocks along the western Kaapvaal Craton margin (Cornell *et al.*, 1998; Moen, 1999; Tinker *et al.*, 2002). The Hartley Basalt Formation within the Olifantshoek succession has yielded a $^{207}Pb/^{206}Pb$ zircon evaporation date of 1928 ± 4 Ma (Cornell *et al.*, 1998). Recent work suggests that pre-Hartley strata previously included in the Olifantshoek Supergroup instead correlate with the older Transvaal Supergroup (Beukes *et al.*, 2002; Evans *et al.*, 2002), which would mean that there are no Olifantshoek strata equivalent in age to pre-1927 Ma strata in the Waterberg Group (*e.g.* Mannyelanong Hill Formation). Correlation of younger parts of the Waterberg succession with Olifantshoek strata overlying the Hartley basalts remains a possibility, however, based on the available age constraints.

In Botswana, the Waterberg Group has been correlated with the Palapye Group exposed farther to the north (*e.g.* Crockett and Jones, 1975; Carney *et al.*, 1994; Williamson, 1996). Recent discovery of ~1.6 Ga zircons in tuffaceous strata near the top of the Palapye Group (Mapeo *et al.*, 2000) suggests that at least the upper part of that group significantly postdates Waterberg deposition. The Waterberg Group has also been correlated with the Umkondo Group in eastern Zimbabwe (*e.g.* Jansen, 1981; Tankard *et al.*, 1982). However, strata in the upper part of the latter group are intercalated with basalts that are extrusive equivalents of the widespread Umkondo dolerite sills (Swift, 1962; McElhinny, 1966; Munyanyiwa, 1999), which have yielded U–Pb zircon and baddeleyite crystallization ages of ~1.1 Ga in eastern Zimbabwe (Hanson *et al.*, 1998; Wingate, 2001; Mukwakwami *et al.*, unpublished data). In addition, there is evidence that sediments in the middle part of the Umkondo Group were only partly lithified at the time of sill intrusion (Munyanyiwa, 1997). Thus, at least the middle and upper parts of the Umkondo Group were emplaced slightly prior to and during 1.1 Ga mafic magmatism and are significantly younger than the Waterberg Group. Paleomagnetic evidence presented herein for coeval emplacement of the Soutpansberg mafic rocks and the ~1879 to ~1872 Ma post-Waterberg dolerites also requires re-evaluation of proposed correlations between the Soutpansberg Group and the Palapye or Umkondo Groups (*cf.* Cheney *et al.*, 1990; Carney *et al.*, 1994).

Regional significance of 1.9 Ga post-Waterberg dolerites

The total extent of ~1.93 and ~1.88 to ~1.87 Ga dolerite intrusions in the Waterberg Group is presently unknown, partly because they are lithologically similar to the ~1.1 Ga Umkondo dolerites present in the same region. Based on available geochronological and paleomagnetic data, the ~1.1 Ga dolerites appear to be more abundant than the older dolerites within the Waterberg Group both in Botswana and South Africa (Figure 2; Jones and McElhinny, 1966; Pancake, 2001; Gose *et al.*, 2003). Mafic sills are also widespread in the older Transvaal Supergroup but are generally

lithologically distinct from the post-Waterberg dolerites and are considered for the most part to be pre- or syn-Bushveld in age. However, dolerite intrusions petrographically similar to the post-Waterberg dolerites cut Transvaal strata in places (Cawthorn *et al.*, 1981; Sharpe, 1981; 1984) and probably relate either to the ~1.1 Ga Umkondo Province or to the ~1.9 Ga intrusions documented herein. Relations between the ~1.93 and ~1.88 to ~1.87 Ga post-Waterberg dolerites and Paleoproterozoic mafic suites in other parts of southern Africa are discussed below.

Hartley Basalt Formation and Trompsburg Complex

Within the Olifantshoek Supergroup along the western craton margin, the U–Pb zircon evaporation date of 1928 ± 4 Ma obtained by Cornell *et al.* (1998) for the Hartley Basalt Formation is within error of our U–Pb baddeleyite ages for the Moshaneng Dolerite in southeastern Botswana. The two units also show geochemical similarities, particularly with regard to Ti, Zr, Y and REE contents (Figures 14 and 15b). The Hartley basalts are inferred to have been extruded during craton-margin rifting (*e.g.* Cornell *et al.*, 1998; Tinker *et al.*, 2002), and the similarities in age and geochemistry raise the possibility that the Moshaneng Dolerite is an intracratonic expression of the same event (*cf.* Cornell *et al.*, 1998). This hypothesis needs to be tested by searching for other mafic rocks of the same age in the largely covered region between the craton margin and the Moshaneng area in southeastern Botswana.

The error circles of poles JP9, JP10 and A1 for the Moshaneng Dolerite do not overlap the error circle of the pole for the Hartley basalts (Figure 12). Evans *et al.* (2002) tentatively inferred that the Hartley pole records a primary thermoremanent magnetization, but petrographic descriptions of the basalts by Cornell (1987) indicate that the rocks have been subjected to low-grade metamorphism. We suggest that the Hartley pole may record remagnetization during post-Hartley contractional deformation that affected this part of the craton margin (Tinker *et al.*, 2002).

Maier *et al.* (2003) recently reported a U–Pb zircon crystallization age of 1915 ± 6 Ma for the Trompsburg Complex, which is located beneath Karoo cover in the southern part of the Kaapvaal Craton (Figure 1). Based on geophysical and borehole data, the complex is a layered gabbroic and anorthositic intrusion with a lopolithic form and a radius of ~25km (Ortlepp, 1959; Buchmann, 1960). The general similarity in the crystallization age of the complex to the U–Pb ages for the Hartley basalts and the Moshaneng Dolerite suggests that all three units are parts of a single regional intraplate magmatic episode.

Soutpansberg mafic rocks

Previous geochronological results (of uncertain geological significance) for the voluminous mafic rocks in the Soutpansberg Group include Rb–Sr whole-rock errorchron dates of 1769 ± 34 Ma for Sibasa basalt and 1749 ± 104 Ma for one of the dolerite sills (Barton, 1979). Our paleomagnetic data from the Soutpansberg mafic rocks show that they were emplaced in the same time frame as intrusion of the ~1.88 to ~1.87 Ga post-Waterberg dolerites, supporting previous interpretations that mafic magmatism in the two areas is directly related (Brandl, 1986; Bumby *et al.*, 2001b). We conclude that the ~1.88 to ~1.87 Ga post-Waterberg dolerites are part of a larger magmatic episode associated with development of the Soutpansberg rift basin.

Geochemical data for the Sibasa basalts and ~1.88 to ~1.87 Ga post-Waterberg sills are compared in Figures 13 to 15, which also include data for dolerite dykes that cut the Waterberg Group, Blouberg Formation and underlying Limpopo basement in the Blouberg area and are inferred by Bumby *et al.* (2001b) to relate to the nearby Sibasa basalts. The basalts and dykes show a wide range of compositions that only partly overlaps with data from the post-Waterberg sills (Figures 13 and 14). REE contents are higher than in the post-Waterberg sills, and the REE patterns are moderately upward-concave, with a mild negative Eu anomaly (Figure 15b; see also Crow and Condie, 1990). These geochemical differences indicate that the post-Waterberg sills did not simply form by lateral injection of Soutpansberg magmas into adjacent parts of the Waterberg Group and instead represent a petrogenetically distinct magma type produced during the same regional magmatic episode.

Mashonaland and Sebanga dolerites

The Mashonaland dolerite sheets intrude Archean basement in the northeastern part of the Zimbabwe Craton (Figure 1). The Sebanga dykes extend over larger parts of the craton and have been inferred to relate to the Mashonaland dolerites (Wilson *et al.*, 1987; Wilson, 1990), although they yield different poles (Figure 12; Bates and Jones, 1996). No trace-element data are available for the Sebanga dykes. The Mashonaland dolerites show a wide range in composition that generally overlaps with data for the 1.93 and ~1.88 to ~1.87 Ga post-Waterberg dolerites (Figure 13). REE profiles for the Mashonaland dolerites are similar to the profile for the ~1.93 Ga Moshaneng Dolerite but are clearly different from the profiles for the ~1.88 to ~1.87 Ga post-Waterberg dolerites, with less enrichment in the LREE and a tendency to develop negative Eu anomalies (Figure 15d). The Mashonaland dolerites have yielded Rb–Sr isotopic data inferred by Compston and McElhinny (1975) and Cahen *et al.* (1984) to constrain crystallization at ~1.9 to ~1.8 Ga. These Rb–Sr results from ancient mafic rocks are highly suspect, and robust geochronological constraints are needed for the Mashonaland dolerites. However, the Mashonaland pole is clearly distinct from our poles for the ~1.93 and ~1.88 to ~1.87 Ga post-Waterberg dolerites (Figure 12), suggesting that the Mashonaland dolerites do not correlate directly with either of these two suites.

Bates and Jones (1996) obtained two different poles for the Sebanga dykes. One pole (Seb-1 in Figure 12) is similar to the JP9, JP10 and A1 poles from the ~1.93 Ga Moshaneng Dolerite. Although data from more sites would be desirable from the Moshaneng area, the present data are consistent with emplacement of some of the Sebanga dykes in the same time frame as the Moshaneng Dolerite. The other Sebanga pole (Seb-2) is statistically indistinguishable from our WSD pole (Figure 12). This result suggests that other members of the Sebanga dyke suite in Zimbabwe are coeval with the ~1.88 to ~1.87 Ga post-Waterberg dolerites and Soutpansberg mafic rocks in South Africa.

Summary

The present data delineate two temporally distinct post-Bushveld Paleoproterozoic intraplate magmatic episodes within the Kaapvaal Craton and along its northern and western margins. Magmatism at ~1928 to ~1915 Ma is recorded by the Hartley basalts along the western margin (Cornell *et al*., 1998), the Trompsburg complex in the southern part of the craton (Maier *et al*., 2003) and the Moshaneng Dolerite in Botswana, ~500 km to the north (this paper). The similarity in paleomagnetic pole positions for the Moshaneng Dolerite and some of the Sebanga dykes suggests that this igneous event may also extend north into the Zimbabwe Craton.

Younger intraplate magmatism is represented by ~1879 to ~1872 Ma dolerite sills in the Waterberg Group and by voluminous mafic magmatism in the Soutpansberg rift basin along the northern Kaapvaal Craton margin. Based on paleomagnetic evidence, some of the Sebanga dykes in Zimbabwe were also emplaced during this second episode. Collectively, these data suggest that Paleoproterozoic intraplate magmatism may be more widespread in southern Africa than previously realized, as also pointed out by Maier *et al*. (2003). Precise geochronological data from the Sebanga dykes, and from other examples of intraplate igneous rocks that may have been emplaced in the same time frame (*e.g.* Layer *et al*., 1998), are needed in order to better constrain the spatial and temporal distribution of Paleoproterozoic intraplate magmatism in the region.

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References

Aldiss, D.T., Tombale, A.R., Mapeo, R.B.M. and Chiepe, M. (1989). The geology of the Kanye area. *Botswana Geological Survey Bulletin*, **33**, 170 pp.

Barton, J.M., Jr. (1979). The chemical compositions, Rb–Sr isotopic systematics and tectonic setting of certain post-kinematic mafic igneous rocks, Limpopo Mobile Belt, southern Africa. *Precambrian Research*, **9**, 57-80.

Barton, J.M., Jr. and van Reenen, D.D. (1992). The significance of Rb–Sr ages of biotite and phlogopite for the thermal history of the Central and Southern Marginal Zones of the Limpopo Belt of southern Africa and the adjacent portions of the Kaapvaal Craton. *Precambrian Research*, **55**, 17-31.

Bates, M.P. and Jones, D.L. (1996). A palaeomagnetic investigation of the Mashonaland dolerites, north-east Zimbabwe. *Geophysical Journal International*, **126**, 513-524.

Beukes, N.J., Dorland, H., Gutzmer, J., Nedachi, M. and Ohmoto, H. (2002). Tropical laterites, life on land, and the history of atmospheric oxygen in the Paleoproterozoic. *Geology*, **30**, 491-494.

Bowring, S.A., Erwin, D.H., Jin, Y.G., Martin, M.W., Davidek, K. and Wang, W. (1998). Geochronology of the end-Permian mass extinction. *Science*, **280**, 1039-1045.

Brandl, G. (1981). Geological Map of the Messina Area (Sheet 2230), 1:250,000. *South Africa Geological Survey.*

Brandl, G. (1985). Geological Map of the Pietersburg Area (Sheet 2328), 1:250,000. *South Africa Geological Survey.*

Brandl, G. (1986). The Geology of the Pietersburg Area (Explanation of Sheet 2328). *South Africa Geological Survey*, 43 pp.

Brandl, G. (1987). The Geology of the Tzaneen Area (Explanation of Sheet 2330). *South Africa Geological Survey,* 55 pp.

Brandl, G. (1993). Geological Map of the Ellisras Area (Sheet 2326), 1:250,000. *South Africa Geological Survey.*

Brandl, G. (1996). The Geology of the Ellisras Area (Explanation of Sheet 2326). *South Africa Geological Survey*, 49 pp.

Brandl, G. (2001). The Limpopo Belt and its Environs (Explanation: Geological Map of the Limpopo Belt and its Environs, 1:500,000). *Council for Geoscience, South Africa*, 46 pp.

Buchmann, J.P. (1960). Exploration of a geophysical anomaly at Trompsburg, Orange Free State, South Africa. *Transactions of the Geological Society of South Africa*, **63**, 1-10.

Buick, I.S., Maas, R. and Gibson, R. (2001). Precise U–Pb titanite age constraints on the emplacement of the Bushveld Complex, South Africa. *Journal of the Geological Society of London*, **158**, 3-6.

Buick, I.S., Williams, I.S., Gibson, R.L., Cartwright, I. and Miller, J.A. (2003). Carbon and U–Pb evidence for a Palaeoproterozoic crustal component in the Central Zone of the Limpopo Belt, South Africa. *Journal of the Geological Society of London*, **160**, 601-612.

Bumby, A.J., Eriksson, P.G., van der Merwe, R. and Brümmer, J.J. (2001a). Shear-zone controlled basins in the Blouberg area, Northern Province, South Africa: syn- and post-tectonic sedimentation relating to ca. 2.0 Ga reactivation of the Limpopo Belt. *Journal of African Earth Sciences*, **33**, 445-461.

Bumby, A.J., Eriksson, P.G., van der Merwe, R. and Maier, W.D. (2001b). The stratigraphic relationship between the Waterberg and Soutpansberg Groups in Northern Province, South Africa: evidence from the Blouberg area. *South African Journal of Geology*, **104**, 205-216.

Bumby, A.J., Eriksson, P.G., van der Merwe, R. and Steyn, G.L. (2002). A half-graben setting for the Proterozoic Soutpansberg Group (South Africa): evidence from the Blouberg area. *Sedimentary Geology*, **147**, 37-56.

Cahen, L., Snelling, N.J., Delhal, J. and Vail, J.R. (1984). *The Geochronology and Evolution of Africa.* Clarendon Press, Oxford, 512 pp.

Callaghan, C.C. (1993). The geology of the Waterberg Group in the southern portion of the Waterberg basin. *South Africa Geological Survey Bulletin*, **104**, 83 pp.

Callaghan, C.C., Eriksson, P.G. and Snyman, C.P. (1991). The sedimentology of the Waterberg Group in the Transvaal, South Africa: an overview.

Journal of African Earth Sciences, **13**, 121-139.

Carney, J.N., Aldiss, D.T. and Lock, N.P. (1994). The geology of Botswana. *Botswana Geological Survey Bulletin*, **37**, 113 pp.

Cawthorn, R.G., Davies, G., Clubley-Armstrong, A. and McCarthy, T.S. (1981). Sills associated with the Bushveld Complex, South Africa: an estimate of the parental magma composition. *Lithos*, **14**, 1-16.

Chavagnac, V., Kramers, J.D., Nägler, T.F. and Holzer, L. (2001). The behaviour of Nd and Pb isotopes during 2.0 Ga migmatization in paragneisses of the Central Zone of the Limpopo Belt (South Africa and Botswana). *Precambrian Research*, **112**, 51-86.

Cheney, E.S. and Twist, D. (1986). The Waterberg "basin" — a reappraisal. *Transactions of the Geological Society of South Africa*, **89**, 353-360.

Cheney, E.S., Barton, J.M., Jr. and Brandl, G. (1990). Extent and age of the Soutpansberg sequences of southern Africa. *South African Journal of Geology*, **93**, 664-675.

Coertze, F.J., Jansen, H. and Walraven, F. (1977). The transition from the Transvaal Sequence to the Waterberg Group. *Transactions of the Geological Society of South Africa*, **80**, 145-156.

Compston, W. and McElhinny, M.W. (1975). The Rb–Sr age of the Mashonaland dolerites of Rhodesia and its significance for palaeomagnetic correlation in southern Africa. *Precambrian Research*, **2**, 305-315.

Cornell, D.H. (1987). A field and petrographic study of the Hartley Basalt Formation, Northern Cape Province of South Africa. *Transactions of the Geological Society of South Africa*, **90**, 7-24.

Cornell, D.H., Armstrong, R.A., and Walraven, F. (1998). Geochronology of the Proterozoic Hartley Basalt Formation, South Africa: constraints on the Kheis tectogenesis and the Kaapvaal Craton's earliest Wilson Cycle. *Journal of African Earth Sciences*, **26**, 5-27.

Crockett, R.N. and Jones, M.T. (1975). Some aspects of the geology of the Waterberg System in eastern Botswana. *Transactions of the Geological Society of South Africa*, **78**, 1-10.

Crow, C. and Condie, K.C. (1990). Geochemistry and origin of early Proterozoic volcanic rocks from the Transvaal and Soutpansberg successions, South Africa. *Precambrian Research*, **47**, 17-26.

de Waal, S.A. and Armstrong, R.A. (2000). The age of the Marble Hall diorite, its relationship to the Uitkomst Complex, and evidence for a new magma type associated with the Bushveld igneous event. *South African Journal of Geology*, **103**, 128-140.

du Plessis, C.P. (1987). New perspectives on early Waterberg Group sedimentation from the Gatkop area, northwestern Transvaal. *South African Journal of Geology*, **90**, 395-408.

du Plessis, M.D. (1978). Geological Map of the Nylstroom Area (Sheet 2428), 1:250,000. *South Africa Geological Survey*.

du Toit, A.L. (1939). *The Geology of South Africa*, 2nd Ed. Oliver and Boyd, Edinburgh, 539 pp.

Eglington, B.M. and Armstrong, R.A. (2004). The Kaapvaal Craton and adjacent orogens, southern Africa: a geochronological database and overview of the geological development of the craton. *South African Journal of Geology*, **107**, 13-32.

Emerman, S.H. (1991). Correlation of a dyke swarm in southeastern Botswana with the Pilanesberg dyke swarm, South Africa. *Journal of African Earth Sciences*, **12**, 525-531.

Evans, D.A.D., Beukes, N.J. and Kirschvink, J.L. (2002). Paleomagnetism of a lateritic paleoweathering horizon and overlying Paleoproterozoic red beds from South Africa: Implications for the Kaapvaal apparent polar wander path and a confirmation of atmospheric oxygen enrichment. *Journal of Geophysical Research*, **107**, no. B12, EPM2, 1-22.

French, J.E., Heaman, L.M. and Chacko, T. (2002). Feasibility of chemical U–Th–total Pb baddeleyite dating by electron microprobe. *Chemical Geology*, **188**, 85-104.

Good, N. and de Wit, M.J. (1997). The Thabazimbi–Murchison Lineament of the Kaapvaal Craton, South Africa: 2700 Ma of episodic deformation. *Journal of the Geological Society of London*, **154**, 93-97.

Gose, W., Hanson, R., Pancake, J., Crowley, J., Ramezani, J., Bowring, S., Blenkinsop, T., Mukwakwami, J. and Dalziel, I. (2003). New paleomagnetic and geochronological data from the Mesoproterozoic Umkondo dolerites, southern Africa. *Abstracts of the European Geophysical Union Meeting, Nice, France, Geophysics Research Abstracts,* **5.** 07201.

Hanson, R.E. (2003). Proterozoic geochronology and tectonic evolution of southern Africa. *In:* M. Yoshida, B. Windley and S. Dasgupta (Editors), Proterozoic East Gondwana: Supercontinent Assembly and Breakup. *Geological Society of London Special Publication*, **206**, 428-463.

Hanson, R.E., Martin, M.W., Bowring, S.A. and Munyanyiwa, H. (1998). U–Pb zircon age for the Umkondo dolerites, eastern Zimbabwe: 1.1 Ga large igneous province in southern Africa/East Antarctica and possible Rodinia correlations. *Geology*, **26**, 1143-1146.

Hattingh, P.J. (1986a). The palaeomagnetism of the Merensky Reef footwall rocks of the Bushveld Complex. *Transactions of the Geological Society of South Africa*, **89**, 1-8.

Hattingh, P.J. (1986b). The palaeomagnetism of the main zone of the eastern Bushveld Complex. *Tectonophysics*, **124**, 271-295.

Hattingh, P.J. (1986c). The palaeomagnetism of the main zone of the western Bushveld Complex. *Earth and Planetary Science Letters*, **79**, 441-452.

Hattingh, P.J. (1989). The palaeomagnetism of the upper zone of the Bushveld Complex. *Tectonophysics*, **165**, 131-142.

Hattingh, P.J. and Pauls, N.D. (1994). New palaeomagnetic results from the northern Bushveld Complex of South Africa. *Precambrian Research*, **69**, 229-240.

Holzer, L., Barton, J.M., Jr., Paya, B.K. and Kramers, J.D. (1999). Tectonothermal history of the western part of the Limpopo Belt: tectonic models and new perspectives. *Journal of African Earth Sciences*, **28**, 383-402.

Holzer, L., Frei, R., Barton, J.M., Jr. and Kramers, J.D. (1998). Unraveling the record of successive high grade events in the Central Zone of the Limpopo belt using Pb single phase dating of metamorphic minerals. *Precambrian Research*, **87**, 87-115.

Irvine, T.N. and Baragar, W.R.A. (1971). A guide to the chemical classification of the common volcanic rocks. *Canadian Journal of Earth Sciences*, **8**, 523-548.

Jaeckel, P., Kröner, A., Kamo, S.L., Brandl, G. and Wendt, J.I. (1997). Late Archaean to Early Proterozoic granitoid magmatism and high-grade metamorphism in the central Limpopo Belt, South Africa. *Journal of the Geological Society of London*, **154**, 25-44.

Jansen, H. (1976). The Waterberg and Soutpansberg Groups in the Blouberg area, northern Transvaal. *Transactions of the Geological Society of South Africa*, **79**, 281-291.

Jansen, H. (1981). The Waterberg and Soutpansberg Groups. *In:* Hunter, D.R. (Ed.), *Precambrian of the Southern Hemisphere*. Elsevier, Amsterdam, 536-556.

Jansen, H. (1982). The geology of the Waterberg basins in the Transvaal, Republic of South Africa. *South Africa Geological Survey Memoir*, **71**, 98 pp.

Jones, D.L. and McElhinny, M.W. (1966). Paleomagnetic correlation of basic intrusions in the Precambrian of southern Africa. *Journal of Geophysical Research*, **71**, 543-552.

Kamber, B.S., Kramers, J.D., Napier, R., Cliff, R.A. and Rollinson, H.R. (1995). The Triangle Shear zone, Zimbabwe, revisited: new data document an important event at 2.0 Ga in the Limpopo Belt. *Precambrian Research*, **70**, 191-213.

Key, R.M. (1983). The geology of the area around Gaborone and Lobatse, Kweneng, Kgatleng, Southern and South East Districts. *Botswana Geological Survey District Memoir*, **5**, 230 pp.

Key, R.M. and Ayres, N. (2000). The 1998 edition of the National Geological Map of Botswana. *Journal of African Earth Sciences*, **30**, 427-451.

Key, R.M. and Mothibi, D. (1999). The National Geological Map of the Republic of Botswana, 1:1,500,000. *Botswana Geological Survey*.

Keyser, N. (1997). Geological Map of the Republic of South Africa and the Kingdoms of Lesotho and Swaziland, 1:1,000,000. *Council for Geoscience, South Africa*.

Kirschvink, J.L. (1980). The least-squares line and plane and the analysis of palaeomagnetic data. *Royal Astronomical Society Geophysical Journal*, **62**, 699-718.

Kröner, A., Jaeckel, P., Brandl, G., Nemchin, A.A. and Pidgeon, R.T. (1999). Single zircon ages for granitoid gneisses in the Central Zone of the Limpopo Belt, Southern Africa and geodynamic significance. *Precambrian Research*, **93**, 299-337.

Layer, P.W., Lopez-Martinez, M., Kröner, A., York, D. and McWilliams, M. (1998). Thermochronometry and palaeomagnetism of the Archaean Nelshoogte Pluton, South Africa. *Geophysical Journal International*, **135**, 129-145.

Le Maitre, R.W. (Editor) (1989). A Classification of Igneous Rocks and Glossary of Terms. Recommendations of the International Union of

Geological Sciences Subcommission on the Systematics of Igneous Rocks. *Blackwell Scientific, Oxford, U.K.*, 193 pp.

Ludwig, K.R. (1980). Calculation of uncertainties of U–Pb isotopic data. *Earth and Planetary Science Letters*, **46**, 212-220.

Maier, W.D., Peltonen, P., Grantham, G. and Mänttäri, I. (2003). A new 1.9 Ga age for the Trompsburg intrusion, South Africa. *Earth and Planetary Science Letters*, **212**, 351-360.

Mapeo, R.B.M., Ramokate, L.V., Kampunzu, A.B. and Armstrong, R.A. (2000). The age of the Palapye Group in northeast Botswana and its geological implications. *Journal of African Earth Sciences, 18th Colloquium of African Geology Special Abstracts Issue*, **30**, 58-59.

McCourt, S. and Armstrong, R.A. (1998). SHRIMP U–Pb zircon geochronology of granites from the Central Zone, Limpopo Belt, southern Africa: implications for the age of the Limpopo Orogeny. *South African Journal of Geology*, **101**, 329-338.

McElhinny, M.W. (1966). The paleomagnetism of the Umkondo lavas, eastern Southern Rhodesia. *Royal Astronomical Society Geophysical Journal*, **10**, 375-381.

Meinster, B. (1977). Discussion on H. Jansen: The Waterberg and Soutpansberg Groups in the Blouberg area, northern Transvaal. *Transactions of the Geological Society of South Africa*, **80**, 289-298.

Moen, H.F.G. (1999). The Kheis Tectonic Subprovince, southern Africa: a lithostratigraphic perspective. *South African Journal of Geology*, **102**, 27-42.

Morgan, G.E. (1985). The paleomagnetism and cooling history of metamorphic and igneous rocks from the Limpopo Mobile Belt, southern Africa. *Geological Society of America Bulletin*, **96**, 663-675.

Morgan, G.E. and Briden, J.C. (1981). Aspects of Precambrian palaeomagnetism, with new data from the Limpopo mobile belt and Kaapvaal craton in southern Africa. *Physics of the Earth and Planetary Interiors*, **24**, 142-168.

Munyanyiwa, H. (1997). The Mozambique Belt and Umkondo Group in eastern Zimbabwe. *In:* Conference on Intraplate Magmatism and Tectonics of Southern Africa (Harare, Zimbabwe, 13-17 September), Fieldtrip Guide: The Mozambique Belt of Zimbabwe, *Geology Department, University of Zimbabwe*, 5-16.

Munyanyiwa, H. (1999). Geochemical study of the Umkondo dolerites and lavas in Chimanimani and Chipinge Districts (eastern Zimbabwe) and their regional implications. *Journal of African Earth Sciences*, **28**, 349-365.

Ortlepp, R.J. (1959). A pre-Karoo igneous complex at Trompsburg, Orange Free State, revealed by drilling exploration. *Transactions of the Geological Society of South Africa*, **62**, 33-57.

Pancake, J.A. (2001). *Geochronological and Paleomagnetic Studies of Mesoproterozoic Mafic Igneous Rocks in Botswana.* Unpublished M.S. thesis, Texas Christian University, Texas, USA, 163 pp.

Pearce, J.A. and Cann, J.R. (1973). Tectonic setting of basic volcanic rocks determined using trace element analyses. *Earth and Planetary Science Letters*, **19**, 290-300.

Pearce, J.A. and Norry, M.J. (1979). Petrogenetic implications of Ti, Zr, Y, and Nb variations in volcanic rocks. *Contributions to Mineralogy and Petrology*, **69**, 33-47.

Reischmann, T. (1995). Precise U/Pb age determination with baddeleyite (ZrO_2), a case study from the Phalaborwa Igneous Complex, South Africa. *South African Journal of Geology*, **98**, 1-4.

SACS (1980). Stratigraphy of South Africa, Part 1. *South Africa Geological Survey Handbook*, **8**, 690 pp.

Schaller, M., Steiner, O., Studer, I., Holzer, L., Herwegh, M. and Kramers, J.D. (1999). Exhumation of Limpopo Central Zone granulites and dextral continent-scale transcurrent movement at 2.0 Ga along the Palala Shear Zone, Northern Province, South Africa. *Precambrian Research*, **96**, 263-288.

Sharpe, M.R. (1981). Petrology and geochemistry of pre-Bushveld and Waterberg mafic sills. *Transactions of the Geological Society of South Africa*, **84**, 75-83.

Sharpe, M.R. (1984). Petrography, classification and chronology of mafic sill intrusions beneath the eastern Bushveld Complex. *South Africa Geological Survey Bulletin*, **77**, 40 pp.

Stacey, J.S. and Kramers, J.D. (1975). Approximation of terrestrial lead isotope evolution by a two-stage model. *Earth and Planetary Science Letters*, **26**, 207-221.

Steiger, R.H. and Jäger, E. (1977). Subcommission on geochronology; convention on the use of decay constants in geo- and cosmochronology. *Earth and Planetary Science Letters*, **36**, 359-362.

Strauss, C.A. (1947). The petrology of a small dolerite and granophyre complex in the New Belgium Block, Potgietersrust District. *Transactions of the Geological Society of South Africa*, **50**, 73-101.

Stubbs, H.M., Hall, R.P., Hughes, D.J. and Nesbitt, R.W. (1999). Evidence for a high Mg andesitic parental magma to the East and West satellite dykes of the Great Dyke, Zimbabwe: a comparison with the continental tholeiitic Mashonaland sills. *Journal of African Earth Sciences*, **28**, 325-336.

Sun, S.-s. and McDonough, W.F. (1989). Chemical and isotopic systematics of oceanic basalts: implications for mantle compositions and processes. *In:* Saunders, A.D. and Norry, M.J. (Eds.), Magmatism in the Ocean Basins. *Geological Society of London Special Publication*, **42**, 313-345.

Swift, W.H. (1962). The geology of the Middle Sabi Valley. *Southern Rhodesia Geological Survey Bulletin*, **52**, 30 pp.

Tankard, A.J., Jackson, M.P.A., Eriksson, K.A., Hobday, D.K., Hunter, D.R. and Minter, W.E.L. (1982). *Crustal Evolution of Southern Africa: 3.8 Billion Years of Earth History*, Springer-Verlag, New York, 523 pp.

Tinker, J., de Wit, M. and Grotzinger, J. (2002). Seismic stratigraphic constraints on Neoarchean–Paleoproterozoic evolution of the western margin of the Kaapvaal Craton, South Africa. *South African Journal of Geology*, **105**, 107-134.

van der Neut, M., Eriksson, P.G. and Callaghan, C.C. (1991). Distal alluvial fan sediments in early Proterozoic red beds of the Wilgerivier Formation, Waterberg Group, South Africa. *Journal of African Earth Sciences*, **12**, 537-547.

Walraven, F. (1997). Geochronology of the Rooiberg Group, Transvaal Supergroup, South Africa. *University of the Witwatersrand, Economic Geology Research Unit, Information Circular, Johannesburg, South Africa*, **316**, 21 pp.

Walraven, F. and Hattingh, E. (1993). Geochronology of the Nebo Granite, Bushveld Complex. *South African Journal of Geology*, **96**, 31-41.

Walraven, F., Armstrong, R.A. and Kruger, F.J. (1990). A chronostratigraphic framework for the north-central Kaapvaal craton, the Bushveld Complex and the Vredefort structure. *Tectonophysics*, **171**, 23-48.

Wang, P. and Glover, L. (1992). A tectonics test of the most commonly used geochemical discriminant diagrams and patterns. *Earth-Science Reviews*, **33**, 111-131.

Ward, S.E. (2002). *Petrogenesis of Mesoproterozoic mafic dykes and sills in Zimbabwe.* Ph.D. thesis (unpublished), University of Portsmouth, Portsmouth, UK, 350 pp.

Williamson, I.T. (1996). The geology of the area around Mmamabula and Dibete. *Botswana Geological Survey District Memoir*, **6**, 239 pp.

Wilson, J.F. (1990). A craton and its cracks: some of the behaviour of the Zimbabwe block from the Late Archaean to the Mesozoic in response to horizontal movements, and the significance of some of its mafic dyke fracture patterns. *Journal of African Earth Sciences*, **10**, 483-501.

Wilson, J.F., Jones, D.L. and Kramers, J.D. (1987). Mafic dyke swarms in Zimbabwe. *In:* Halls, H.C. and Fahrig, W.F. (Eds.), Mafic Dyke Swarms. *Geological Association of Canada Special Paper*, **34**, 433-444.

Winchester, J.A. and Floyd, P.A. (1977). Geochemical discrimination of different magma series and their differentiation products using immobile elements. *Chemical Geology*, **20**, 325-343.

Wingate, M.T.D. (2001). SHRIMP baddeleyite age for an Umkondo dolerite sill, Nyanga Mountains, eastern Zimbabwe. *South African Journal of Geology*, **104**, 13-22.

Zimbabwe Geological Survey (1994). Geological Map of Zimbabwe, 1:1,000,000. *Zimbabwe Geological Survey.*

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